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Deskilling, reskilling, or upskilling? Unpacking the pathways of student adaptation to generative artificial intelligence

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A B S T R A C T

The proliferation of generative AI (GAI) like ChatGPT is transforming how students engage with information and knowledge-focused activities in higher education, sparking debate about its dual impact on learning. While GAI offers potential benefits like enhanced efficiency, concerns about risks such as skill erosion persist. To address this tension, we investigate how students' dependence on GAI shapes their learning outcomes through skill adaptation processes and under what conditions these effects occur. We conducted a three-phase mixed-methods study (survey N = 306; interviews N = 16; experiment N = 397). Our findings reveal that GAI dependence, influenced by individual learning goals (performance-avoidance/-approach), drives three distinct skill adaptation processes: deskilling (skill erosion), reskilling (acquiring new GAI-related competencies), and upskilling (enhancing existing skills). These adaptations, in turn, differentially impact routine and innovative performance. Qualitative results corroborate and complement these findings, indicating that task characteristics shape GAI use patterns into substitutive and augmentative use. Finally, a scenario-based experiment provides causal evidence for this emergent insight, demonstrating how task characteristics drive the adoption of substitutive vs. augmentative use, which in turn leads to divergent skill adaptation pathways. By combining diverse methodologies, this study clarifies the lights and shadows of GAI dependence, demonstrating how its effects are contingent on individual agency, technological appropriation (substitutive vs. augmentative), and task context. Our findings advance theory on human-AI adaptation and provide practical guidance for practitioners to optimize GAI's role in learning and knowledge-focused activities.

1. Introduction

Compared to earlier generations of digital technologies, generative artificial intelligence (GAI) tools like ChatGPT have exerted a more transformative impact in education, changing the ways students learn (O'Dea, 2024). Beyond education, GAI represents a powerful wave of information technology (IT) that is reshaping knowledge work (Benbya et al., 2024). Understanding how individuals adapt to such intelligent tools is therefore a central concern for information management, as GAI fundamentally alters how people process information and create knowledge. GAI empowers students to efficiently undertake various tasks, including acquiring knowledge, coding software, and composing textual content, through natural language interaction (Lee et al., 2024). According to a report, the AI education market is expected to grow from 3.25 billion USD in 2023 to 53.11 billion USD by 2032 (Shahzad et al., 2024). GAI tools like ChatGPT are prominent examples of AI applications in education and have attracted significant attention from educators and students since their launch (Larson et al., 2024). A large-scale study, for instance, shows that more than half of higher education students regularly use ChatGPT in their academic activities (Stöhr et al., 2024). This high adoption has led to a growing dependence on GAI among students (Ye et al., 2024; Zhang et al., 2024).

Despite the prevalence of GAI use, controversy exists regarding its role in managing learning processes and shaping performance outcomes. This debate mirrors the broader discourse on the "lights and shadows" of GAI. On the one hand, GAI tools are lauded for their potential to enhance efficiency, provide personalized support, and empower students with new capabilities (Casaló et al., 2025; Dwivedi et al., 2023; Sun et al., 2024). On the other hand, serious concerns are raised, including ethical issues (Belk et al., 2023), psychological tensions (Flavián et al., 2024), over-dependence leading to superficial learning (Feng et al., 2024), and the erosion of critical thinking and core skills (Belanche et al., 2024). This tension manifests as a "***productivity paradox***" (Lim et al., 2023): although the intrinsic nature of GAI tools is neutral, students'

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dependence on GAI might yield both positive and negative learning outcomes. Prior studies often adopt a ***technology-determinism*** perspective, focusing on the direct impact of GAI usage on performance, assuming technology directly causes outcomes (Abbas et al., 2024; Rehman et al., 2024). However, this overlooks the significant role of ***student adaptation***. Specifically, students might adapt positively, leveraging GAI for self-regulated learning or knowledge acquisition (Zhou & Kim, 2024), or negatively, leading to laziness or restricted critical thinking (Crawford et al., 2024; Zhang et al., 2024). To better understand the GAI use-performance relationship, it is crucial to consider user adaptation from a ***human-agency*** perspective (Giannakos et al., 2024; Larson et al., 2024; Naamati-Schneider & Alt, 2024). A particularly relevant aspect of this adaptation involves changes in students' skills (French & Shim, 2024; Suriano et al., 2025). Therefore, understanding the productivity paradox necessitates investigating skill adaptation—how students' skill sets evolve in their dependence on GAI—as a central underlying process. Moreover, the key question in understanding this paradox is not just whether dependence impacts performance, but how (skill adaptation) and under what conditions (individual characteristics and contextual factors) these impacts occur. Addressing these questions is critical not only for optimizing educational practice, but also for the broader field of information management concerned with human adaptation to technologies and user delegation experience when interacting with GAI (Adam et al., 2024; Flavián et al., 2024; Rob et al., 2025). Therefore, we employed a sequential mixed-methods approach to investigate the underlying processes and contingencies shaping this relationship, answering the following research questions (RQ):

RQ1. How do students' skill adaptations act as the underlying mechanisms through which dependence on GAI tools influences their learning performance?

RQ2. How do contextual factors (task characteristics) shape students' patterns of GAI dependence (substitutive vs. augmentative use), and how do these patterns subsequently impact skill adaptation?

To address these research questions, we focus on three crucial aspects. First, from a ***human-agency perspective***, we recognize that individuals may deploy different adaptation strategies when faced with new technologies like GAI (Nevo et al., 2021). Yet, previous research has paid little attention to how individual differences shape GAI use (Budhathoki et al., 2024; Gao et al., 2024; Jo, 2024). To fill this gap, our first objective is to identify key skill adaptation processes and explore the impact of individual characteristics (i.e., learning goals) on these processes.

Second, another way to solve the productivity paradox of GAI is to refresh the conceptualization and measurement of learning performance. Prior studies tend to treat learning performance as a general or unified concept without distinguishing its different types, or only focus on one specific type of learning performance, i.e., routine performance (Al-Qaysi et al., 2024; Jo, 2023; Zhou & Kim, 2024). However, unlike routine performance, which usually involves following a structured, task-oriented approach, innovative performance is concerned with the ability to think creatively, explore new possibilities, and generate novel ideas (Shirish et al., 2021). For different types of learning performance, GAI use may exert different impacts. Thus, our second research objective is to differentiate between routine and innovative performance and investigate the differential impacts of the three skill adaptation processes on these distinct outcomes.

Third, fully understanding how GAI dependence influences skill adaptation requires examining the nature of that dependence itself. While dependence is prevalent, the way students rely on GAI—whether primarily to replace their effort or to augment their capabilities—likely varies and carries different implications (Guo et al., 2024; Raisch & Krakowski, 2021). Furthermore, factors like task context are known to influence technology use patterns (e.g., Goodhue & Thompson, 1995), yet the specific interplay between learning task characteristics, the resulting GAI dependence patterns (i.e., substitutive vs. augmentative use), and their downstream effects on skill adaptation remains largely unexplored. Therefore, our third research objective is to investigate how learning task characteristics (i.e., analyzability and variety) influence students' adoption of substitutive vs. augmentative GAI dependence patterns, and how these distinct patterns subsequently shape the processes of skill adaptation.

To fill the mentioned research gaps, this study integrates multiple theoretical perspectives within an overarching adaptive structuration theory (AST) framework (DeSanctis & Poole, 1994; Schmitz et al., 2016). AST provides a lens to understand the relationship between students (agents), GAI technology (structure), and the adaptation processes that shape outcomes (Schmitz et al., 2016). First, we draw on labor process theory (Adler, 2007) to conceptualize the three skill adaptation strategies (***upskilling, deskilling, and reskilling***) and achievement goal theory (Nevo et al., 2021) to understand the role of learning goals. Second, we differentiate between routine and innovative performance and theorize the differential impacts of three kinds of user adaptation on them. Third, we operationalize substitutive and augmentative use based on the automation-augmentation framework (Guo et al., 2024; Raisch & Krakowski, 2021) and examine task characteristics (Daft & Macintosh, 1981) as key contextual antecedents within the AST framework. This integrated theoretical approach is investigated using a sequential mixed-methods design (Quantitative → Qualitative → Experiment). Study 1 (quantitative survey) provided an initial test of the model linking goals, dependence, skill adaptations, and differentiated performance. Study 2 (qualitative interviews) corroborated these findings and explored contextual factors, crucially informing Study 3 by revealing the importance of task characteristics and suggesting the substitutive/augmentative use distinction. Building directly on these insights, Study 3 (experiment) rigorously tested the causal chain (task characteristics → substitutive/augmentative use → skill adaptations) and allowed for a re-examination of skill adaptation effects on performance (Study 1) with enhanced causal clarity. This multi-phase design allows our integrated framework to progressively explore how skill adaptation processes unfold to explain the GAI productivity paradox under the influence of individual goals and specific contextual factors.

2. Literature review

2.1. GAI in learning

The integration of GAI into various societal domains, including education, represents a significant shift in human-AI interaction and information management practices. As a prominent GAI application, ChatGPT presents a significant opportunity to transform educational and learning processes. For students in particular, GAI tools like ChatGPT offer significant value by providing quick responses and personalized learning support (Yan et al., 2023). These capabilities, which can include both analytical and emotional AI functionalities, position GAI as a powerful tool for enhancing learning efficiency (Casaló et al., 2025). For example, GPT-4 has demonstrated high proficiency in argumentative writing tasks (Zhang et al., 2025). This aligns with the potential of AI to create value through enhanced decision support (Wirtz et al., 2018). However, alongside these benefits, significant concerns constituting the "dark side" of GAI have emerged (Belanche et al., 2024; Giannakos et al., 2024). Critics argue that excessive reliance on GAI may undermine the value of personal effort, running contrary to the core objectives of teaching and learning (Giannakos et al., 2024; Yan et al., 2023).

In light of the considerable interest in integrating GAI into learning environments, there have been numerous studies investigating the factors leading to the adoption of GAI tools (Foroughi et al., 2023; Saif et al., 2024). Specifically, drawing upon various technology acceptance theories including TAM and UTAUT, prior studies have explored the impacts of performance expectancy, effort expectancy, social influence, facilitating conditions (Budhathoki et al., 2024), and personal innovativeness on adoption intention (Foroughi et al., 2023; Strzelecki, 2024) or actual usage behavior (Saif et al., 2024). While these adoption studies explore the antecedents of GAI use, they often stop short of examining the post-adoption processes. Specifically, there is less understanding of how students appropriate these tools in their learning activities and the subsequent, often paradoxical, effects on their performance. Another stream of research directly engages with the paradoxical impacts of GAI on learning processes and outcomes. On the one hand, scholars have highlighted its benefits, finding that GAI use can enhance knowledge construction (Wu et al., 2024), improve academic engagement (Rehman et al., 2024), and boost learning performance (Al-Qaysi et al., 2024). On the other hand, some studies point to detrimental effects, linking GAI use to outcomes such as memory loss, procrastination, and restricted creativity (e.g., Zhang et al., 2024). Collectively, these discussions often center on whether GAI empowers learners or replaces critical cognitive functions, reflecting the psychological tensions involved (Flavián et al., 2024). These controversial findings indicate that the role of GAI usage in learning is more complex than simply negative or positive, and call for in-depth understanding of the process through which it exerts its influence (O'Dea, 2024). Given that AST can shed light on the process through which student users adapt to GAI technologies, this theory is taken as the theoretical foundation of the present study.

2.2. Adaptive structuration theory

AST initially focuses on how users in organizations use a particular technology through structured processes, providing a framework for understanding the use of organizational IT and its outcomes (DeSanctis & Poole, 1994). While originally focused on groups, AST has been extended to the individual level as ASTI, making it directly applicable to our study of student learning (Schmitz et al., 2016). Crucially for our research, ASTI posits that individual adaptation is shaped by the confluence of three structural sources: technology characteristics, individual characteristics, and task characteristics. Our research model is built upon this theoretical triad, examining GAI dependence (technology), learning goals (individual), and task context (task) as the key drivers of the adaptation process. Consistent with ASTI's components and the logic of our sequential mixed-methods design, we phase our investigation. We first examine the influence of general technology and individual characteristics in Study 1 to build a foundational model. We then isolate task characteristics for causal testing in our Study 3 experiment, allowing for a further analysis of how task contexts shape adaptation.

ASTI argues that users adapt to the new technology through a social interaction process following the "input-process-output" framework (Schmitz et al., 2016). Specifically, the inputs are the structural antecedents: individual and technology characteristics. The core process is the user adaptation itself. The final outputs are the performance outcomes that emerge from this technology use. However, applying AST/ASTI to the context of intelligent and adaptive GAI tools warrants a critical consideration of its boundaries (Gupta & Bostrom, 2009; Rob et al., 2025; Schmitz et al., 2016). While AST acknowledges user interpretation (e.g., "spirit," "appropriation"), its primary analytical lens often centers on observable interactions and social structures, potentially underemphasizing the detailed cognitive processes—such as shifts in mental models or learning strategies—that are especially pertinent when users adapt to technologies deeply involved in knowledge work (Gupta & Bostrom, 2009; Rob et al., 2025). Recent work highlights this by proposing extensions like "technocognitive structuration" to better integrate cognitive mechanisms into ASTI (Rob et al., 2025).

Therefore, while using ASTI as our foundational framework, we enrich it to capture the cognitive capability shifts central to GAI interaction. We achieve this in three ways. First, we identify three types of skill adaptation, namely deskilling, upskilling, and reskilling (outcomes of the GAI structuration process). Second, technology dependence and individual learning goals are identified as technology and individual characteristics that serve as antecedents to skill adaptation. Furthermore, informed by ASTI's differentiation between exploitative and exploratory adaptation (Rob et al., 2025; Schmitz et al., 2016) and emergent qualitative findings, we later investigate distinct GAI use patterns (substitutive vs. augmentative) as appropriation modes reflecting potentially different cognitive engagements. Third, we demonstrate how these appropriation modes are causally shaped by task context, thereby providing a complete AST-based model of student adaptation to GAI.

2.3. Skill adaptation: deskilling, upskilling and reskilling

A central tension in the age of GAI lies in its multifaceted impact on individual skills, a crucial factor in understanding how GAI may lead students to feel their capabilities are being augmented or substituted (Belanche et al., 2024; Flavián et al., 2024). To conceptualize these capability shifts, we draw on labor process theory (LPT), a key framework for understanding how technology alters individual skills (Adler, 2007). The updated perspective of this theory indicates that the adoption of technology in organizations can lead to not only deskilling but also upskilling and reskilling of the workforce (Dornelles et al., 2023). This process, however, is rarely a simple substitution, but often involves a fundamental repositioning and refocusing of an individual's existing skill set (Xing & Sharif, 2025). Given the intricacy of new technology's application, the reskilling perspective balances the impact of upskilling and deskilling, becoming an important goal for organizations and individuals (Alvarez, 2008). Reskilling for the workforce means that employees need new knowledge and skills to take on different or entirely new roles (Li, 2022). For instance, in the context of digitalization and automation, drivers anticipate a clear shift from traditional mechanical expertise towards new IT-related proficiencies, even as human-centric abilities like customer interaction remain indispensable (Gong et al., 2025). The introduction of technology can enhance workers' qualifications through reskilling, which involves higher complexity and creativity (Dornelles et al., 2023). Crucially, this adaptation is not merely a passive reaction; individuals can play a highly proactive role in reshaping their tasks and skill sets in response to new technology (Li et al., 2024).

Drawing from this broader human-capital perspective on workforce adaptation, we define skill adaptation as the process by which individuals adjust their skills to better align with the demands of a specific task or technology (Schmitz et al., 2016). This process is essential for achieving desired outcomes, especially when engaging with GAI in learning contexts (Lee et al., 2024). In the GAI learning context, this skill adaptation is multifaceted (Acar, 2024; Hashmi & Bal, 2024; Suriano et al., 2025). On the one hand, GAI can simplify tasks like data analysis and translation, reducing the need for prior skills and potentially evoking a sense of replacement (i.e., ***deskilling***) (Flavián et al., 2024; French & Shim, 2024; Raisch & Krakowski, 2021). On the other hand, it can enhance existing competencies, fostering a sense of empowerment (i.e., ***upskilling***) (Leon, 2023). Critically, effective GAI use requires new competencies, such as the ability to engineer prompts (Robertson et al., 2024). The development of such GAI-specific abilities is a process of ***reskilling***, which involves integrating new tool-related skills with existing domain knowledge (Dornelles et al., 2023). Fig. 1 summarizes these three distinct adaptation pathways.

Despite the prevailing view that new technologies can reshape individual skills, several gaps remain. First, existing literature primarily focuses on the relationship between organizational technology changes and employee skills in organizational contexts (Dornelles et al., 2023; Li, 2022; Pedota et al., 2023). Whether these findings can be applied to individual technologies (i.e., ChatGPT) and other research contexts (e.g., learning) still requires further investigation. Second, prior studies

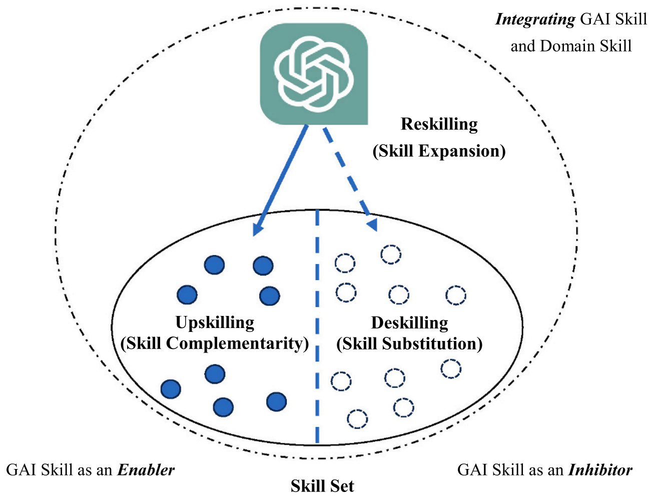

Fig. 1. Impact of GAI tools usage on students' skill set.

have viewed upskilling and deskilling as two opposite ends of one construct, neglecting the fact that new technologies can enhance certain skills while reducing others, which implies different dimensions of skill adaptation that can coexist. Third, measurement instruments for different dimensions of skill adaptation (i.e., deskilling, upskilling, and reskilling) are still lacking and the different impacts of these constructs on learning performance have not been empirically tested (Giannakos et al., 2024).

To address these gaps, we conceptualize and differentiate these three forms of skill adaptation, as summarized in Table 1. Specifically, ***deskilling*** means skill substitution (Dornelles et al., 2023), reflecting the extent to which previous skills are replaced by the new technology. For example, with the support of ChatGPT, it becomes less necessary for students to possess the skill of translation. ***Upskilling*** means skill complementarity (Dornelles et al., 2023), reflecting the extent to which the previously possessed skills are enhanced due to GAI use. For example, GAI enables students to better understand the rationales of knowledge in programming. ***Reskilling*** is defined as the students' ability to utilize GAI for learning, which measures the extent to which they integrate skills related to GAI into their learning (Dornelles et al., 2023). For example, students need to know how to make GAI generate high-quality content through precise instructions and effective multi-turn conversations.

Table 1
Comparison among deskilling, upskilling, and reskilling.

Skill Adaptation	Definitions	Effects	Refs.
Deskilling	**Skill substitution**	**Inhibiting** The reduction in the necessity to learn certain areas of knowledge because GAI tools can perform or simplify tasks previously requiring specific expertise.	Alvarez (2008), Hyatt et al. (2021), Rinta-Kahila et al. (2023), Rowe et al. (2023), Xue et al. (2022)
Upskilling	**Skill complementarity**	**Enabling** The enhancement of existing skills or knowledge using GAI tools, improving the depth or breadth of understanding in a particular area.	Li (2022), Leon (2023), Margherita and Braccini (2021), Pedota et al. (2023), Leon (2023)
Reskilling	**Skill expansion**	**Integrating** Integrating GAI tools with domain knowledge, enabling students to learn new domain knowledge by leveraging GAI tools.	Alvarez (2008), Li (2022), Leon (2023), Tamayo et al. (2023)

2.4. *Learning goals: performance avoidance and approach*

To operationalize the individual characteristics component of the ASTI framework, we turn to achievement goal theory (AGT), which posits that an individual's goals shape their motivation and learning outcomes (Nevo et al., 2021). Within AGT, we focus on two well-established performance goal orientations. A performance-approach goal reflects a desire to demonstrate competence and outperform others, whereas a performance-avoidance goal reflects a desire to avoid performing poorly or appearing incompetent relative to others (Elliot & Harackiewicz, 1996). A vast body of research demonstrates the differential effects of these orientations. Generally, performance-approach goals are linked to positive learning strategies and outcomes, while performance-avoidance goals are predictive of negative academic performance (Darnon et al., 2007, 2009). Furthermore, these goal orientations activate distinct cognitive and behavioral processes (Elliot, 1999). For example, in a technology-mediated context, Nevo et al. (2021) found that performance-approach goals are associated with effort regulation strategies, whereas performance-avoidance goals are not. This evidence, which links goals to specific adaptive behaviors, provides a theoretical basis for hypothesizing how learning goals will influence students' skill adaptation in a GAI-enabled environment.

3. Research model and hypothesis development

Drawing upon AST, we develop an initial nomological network model (see Fig. 2) to explore how technology dependence and learning goals influence three interrelated dimensions of skill adaptation, which in turn affect routine and innovative performance.

3.1. *Learning goals and skill adaptation*

Adaptation is a coping behavior, which is a specific outcome resulting from how individuals deal with IT-related events (Beaudry, 2005; Schmitz et al., 2016). Therefore, different individuals may respond to the challenges brought by IT in different ways (Nevo et al., 2021). Specifically in technology-mediated learning, different types of learning goals have varying impacts on specific coping behaviors and strategies (Janson & Janke, 2024). In this study, we examine two distinct types of learning goals: performance-avoidance, which focuses on avoiding unfavorable evaluations and maintaining a safe status quo, and performance-approach, which focuses on gaining favorable evaluations and performing better than others (Van Laar et al., 2019). Moreover, we argue that these distinct goal orientations will influence which of the three primary skill adaptation strategies—deskilling, reskilling, and upskilling—students adopt in the GAI-enabled learning context.

Deskilling generally refers to the reduction in the need for particular skills due to technological advancements (Dornelles et al., 2023). In the GAI-enabled learning context, deskilling, which involves the redistribution of cognitive load, is a strategy that balances avoiding outcome failure with expending too much effort (Rowe et al., 2023). AGT literature suggests that a performance-avoidance goal can lead to behaviors that minimize efforts in learning (Nevo et al., 2021). Individuals with this goal tend to choose easier paths to completing a task to reduce the risk of failure (Van Laar et al., 2019). GAI tools can help students handle tasks more efficiently by taking over complex problems that traditionally require significant effort and skill (Gao et al., 2024). Therefore, students motivated by a performance-avoidance goal are likely to view GAI as an opportunity to offload cognitive demands, leading them to

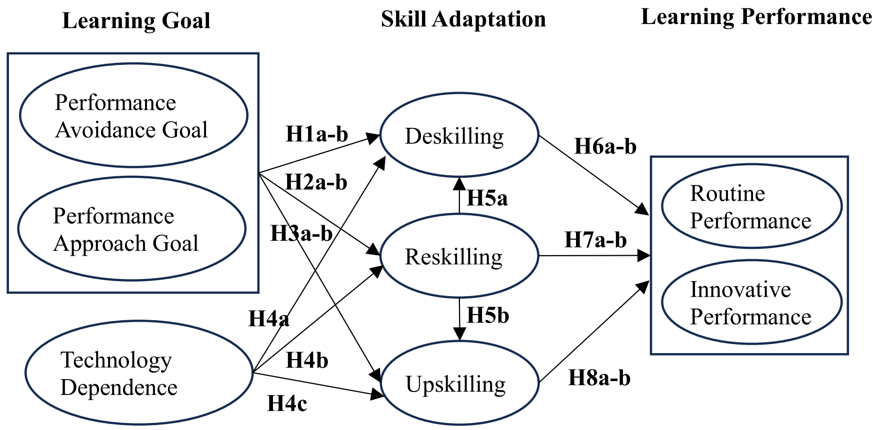

Fig. 2. Initial research model.

rely less on their original skills and thus exacerbating the deskilling process. This aligns with findings that performance-avoidance goals foster surface processing strategies (Nevo et al., 2021), which imply a preference for skill substitution. Thus, we hypothesize:

H1a. Performance-avoidance goal is positively associated with deskilling.

In contrast, performance-approach goals promote engagement in challenging tasks, fostering skill development and mastery (Nevo et al., 2021). Individuals with these goals may utilize technology to complement and expand their skills rather than replace them (Hung et al., 2024). However, the pursuit of superior performance is also a pragmatic one. From a cognitive resource allocation perspective, an approach-oriented student may choose to delegate a task to GAI—thus accepting a degree of deskilling—if that task is not central to the core competencies they aim to showcase. This strategic substitution would free up cognitive resources for more critical activities. Despite this potential contingency, the core drive of a performance-approach goal is to build and demonstrate superior competence (Nevo et al., 2021). Since widespread skill erosion would undermine this primary objective, we expect that, on balance, students with a performance-approach orientation will resist deskilling to protect their overall competitive advantage. Therefore, we hypothesize:

H1b. Performance-approach goal is negatively associated with deskilling.

We now turn to reskilling, which refers to the process of acquiring new skills to adapt to new technological environments (Dornelles et al., 2023). In GAI-enabled learning, reskilling involves students adapting their skill sets and readjusting their capabilities to leverage GAI technologies. AGT posits that performance-avoidance goals can result in a fear of failure and underperformance (Nevo et al., 2021). Students with these goals may fear that using GAI tools will expose their weaknesses with new technology, which in turn can reduce their intrinsic motivation for learning and skill enhancement. Consequently, students with performance-avoidance goals often avoid risk and stick to familiar learning patterns, thus missing opportunities to reskill (Hung et al., 2024). This reduced motivation to learn could hinder them from effectively integrating GAI tools into their learning processes, further inhibiting reskilling. Therefore, we propose:

H2a. Performance-avoidance goal is negatively associated with reskilling.

Conversely, performance-approach goals, which emphasize achieving success and outperforming others, foster a different response (Nevo et al., 2021). Individuals with these goals are more open to embracing change and trying new technology to achieve their goals (Leung et al., 2022). A performance-approach orientation is expected to be positively related to skill acquisition (Van Laar et al., 2019). Empirical evidence on student interactions with new technologies confirms this, indicating that performance-approach goals are linked to greater engagement and more adaptive strategies, such as investing time to learn a technology that is perceived as instrumental to success (Janson & Janke, 2024). Thus, students with performance-approach goals are likely to view learning how to use GAI effectively (i.e., reskilling) as a competitive edge. We hypothesize:

H2b. Performance-approach goal is positively associated with reskilling.

Finally, we consider upskilling, the process by which technology helps individuals enhance their existing skills and better perform their original tasks (Dornelles et al., 2023). In GAI-enabled learning, upskilling refers to the enhancement of existing skills through the integration of GAI tools in the learning process. According to AGT, performance-avoidance goals often lead to a fear of failure and the adoption of superficial learning strategies, which are antithetical to deep skill improvement (Nevo et al., 2021). Students with these goals often expend a significant amount of cognitive resources in managing anxiety, thereby limiting their capacity to improve skills (Darnon et al., 2007). This avoidance behavior is particularly harmful in a technology-mediated learning environment, where engagement with technological tools is critical for upskilling (Söllner et al., 2018). Thus, performance-avoidance goals may prevent students from exploring the potential of GAI tools beyond basic use, limiting their skill development. We propose:

H3a. Performance-avoidance goal is negatively associated with upskilling.

On the contrary, performance-approach goals involve the motivation to outperform others, which can foster greater engagement with the learning process and promote the adoption of effective learning strategies (Cho & Cho, 2014). Students with these goals tend to engage more in challenging learning activities, which are essential for upskilling, leading to sustained effort in skill development (Huang, 2016). Such goals promote active learning behaviors (Van Laar et al., 2019), including seeking feedback, representing knowledge, and regulating effort, which enable students to fully exploit the potential of GAI tools, leading to significant upskilling. Therefore, we hypothesize:

H3b. Performance-approach goal is positively associated with upskilling.

3.2. *Technology dependence and skill adaptation*

Technology dependence refers to the extent to which students rely on

GAI tools in their learning (Shu et al., 2011). We theorize that this technology dependence is a critical driver of adaptation, potentially creating different pathways for skill substitution (deskilling), new skill acquisition (reskilling), and existing skill enhancement (upskilling).

First, dependence on GAI can lead to deskilling by automating tasks that once required the student's own cognitive effort. According to LPT (Adler, 2007), automation technology can reduce the need for specialized human skills, leading to a simplification of labor and potentially reducing the skill levels of the employees. Empirical studies have supported the notion that dependence on technology can lead to deskilling (Dornelles et al., 2023; Rowe et al., 2023), a finding that can be extrapolated to educational environments. In GAI-enabled learning, students' excessive dependence on GAI tools for tasks such as writing and translating might lead to an erosion of their fundamental skills in these domains, causing deskilling (Acar, 2024). Thus, we propose:

H4a. Technology dependence is positively associated with deskilling.

Beyond skill substitution, however, dependence on GAI can also enable students to acquire entirely new competencies, a process known as reskilling. To effectively leverage a technology one depends upon, a user must learn how to interact with it proficiently (Lou et al., 2022; Trieu, 2023). In the GAI context, this implies that for dependent students to achieve high-quality outcomes, they are motivated to learn the nuances of the technology, such as mastering prompt engineering or critically evaluating GAI outputs (Abbas et al., 2024). Their continuous effort to adapt GAI to their needs drives an expansion of their skill sets. Thus, we hypothesize:

H4b. Technology dependence is positively associated with reskilling.

Similarly, dependence on GAI may create opportunities for upskilling, the enhancement of a user's pre-existing competencies (Leon, 2023; Zirar, 2023). This can occur when a student uses GAI as a cognitive scaffold, for instance, by interacting with it to deepen their understanding of a complex topic or by observing how it solves a problem to refine their own analytical reasoning (Suriano et al., 2025; Wang et al., 2024). When GAI is used in this augmentative way, the sustained interaction provides a platform for enhancing existing skills (Alzubi et al., 2025). Based on this potential positive pathway, we propose:

H4c. Technology dependence is positively associated with upskilling.

3.3. The effect of reskilling on deskilling and upskilling

According to AST, technology adoption is regarded as an "appropriation of structure" that plays an active role in shaping social interactions (DeSanctis & Poole, 1994; Schmitz et al., 2016). AST also posits that technology can influence the generation of new structures through interactions among people and technology (DeSanctis & Poole, 1994). In GAI-enabled learning, students need to expand their skill set to adapt to GAI tools, a process known as reskilling. For instance, the ability to write precise prompts to obtain high-quality answers is a newfound skill, emerging with the advent of GAI tools (Robertson et al., 2024). This shift represents a significant change in how students engage with their learning processes. Therefore, we conceptualize reskilling itself as an emergent structure—a distinct mode of appropriation through which students adapt to GAI tools. From an AST perspective, deskilling and upskilling represent changes to a student's pre-existing skill structures. In contrast, reskilling constitutes an emergent structure, one that did not exist before the technology's introduction. A central principle of AST is that such emergent structures, once appropriated by users, can in turn reshape existing social and capability structures (DeSanctis & Poole, 1994). We therefore argue that the appropriation of this new, emergent structure (reskilling) will influence the state of pre-existing skill structures (deskilling and upskilling).

Specifically, in GAI-enabled learning, reskilling transcends mere technology adoption, indicating that students can adjust their skill structures using GAI tools (Giannakos et al., 2024). Reskilling is characterized by active engagement with GAI, which in turn fosters reflective thinking, critical thinking, and self-regulated learning (Suriano et al., 2025). Through this reflective process, students are encouraged to apply their core knowledge to better cope with the challenge of skill replacement. Thus, we propose:

H5a. Reskilling is negatively associated with deskilling.

In addition, the acquisition of new GAI-related skills is also likely to enhance a student's existing competencies. When students are deeply engaged with GAI tools, the mastery of new GAI-related skills can create an augmentative effect that elevates their original capabilities (Li et al., 2025). For instance, a student who masters advanced prompting techniques can use GAI to generate more sophisticated code examples, which in turn deepens their understanding of coding. This creates a positive effect where the new structure (reskilling) positively reinforces the old structure (upskilling) (Schmitz et al., 2016). Thus, we propose:

H5b. Reskilling is positively associated with upskilling.

3.4. Skill adaptation and learning performance

In this study, we identify routine and innovative performance as two dimensions of learning performance. Specifically, we define routine performance as a student's effectiveness and efficiency in completing learning tasks. In contrast, innovative performance refers to a student's ability to generate novel ideas and develop creative strategies in their learning (Shao et al., 2022). We argue that deskilling exerts a divergent influence on the two dimensions of learning performance. On the one hand, by automating routine work and simplifying complex procedures, deskilling can enhance routine performance (Wang et al., 2024). This substitution effect improves efficiency and allows students to complete standard tasks with less effort (e.g., data analysis, translation). Therefore, we propose:

H6a. Deskilling is positively associated with routine performance.

On the other hand, this same process of cognitive offloading may undermine innovative performance, as tasks that once required complex cognitive skills are simplified as GAI tools take over, reducing the need for those skills (Acar, 2024). In this case, students rely more on the technology and less on their own development of critical thinking and creativity, potentially degrading their innovative capabilities (Alzubi et al., 2025). Thus, we propose:

H6b. Deskilling is negatively associated with innovative performance.

In contrast to deskilling, we propose that reskilling offers benefits to both performance dimensions. For routine performance, the core benefit of reskilling lies in the efficient use of the GAI tool itself. As students develop GAI-related skills, such as structuring multi-turn conversations, they become more capable of generating relevant outputs quickly, which leads to greater efficiency. Supporting this, previous research indicates that developing relevant technology skills contributes to improved routine performance through deeper system integration and routinization (Carraher Wolverton et al., 2023). Therefore, we hypothesize:

H7a. Reskilling is positively associated with routine performance.

Beyond efficiency gains, reskilling can also foster innovative performance. Innovation is often the result of combining new ideas (Shao et al., 2022). Through reskilling, students learn to explore the GAI tool in new ways. This exploration exposes them to diverse outputs, supporting the generation of novel ideas and enhancing their creativity (Nevo et al., 2020). Students who effectively integrate technology into their learning are better positioned to harness it for creative ends (Shahzad et al., 2024). Thus, we propose:

H7b. Reskilling is positively associated with innovative performance.

Finally, we argue that upskilling, like reskilling, positively influences both performance dimensions. When students use GAI to enhance their existing competencies (e.g., improving their programming logic), their ability to perform routine tasks that depend on these skills is directly improved. The GAI acts as a cognitive scaffold, enabling a higher level of execution on standard tasks, which leads to better routine performance (Giannakos et al., 2024). Accordingly, we hypothesize:

H8a. Upskilling is positively associated with routine performance.

Moreover, upskilling is essential for innovative performance. Innovation is fundamentally a learning process where new ideas are built upon a foundation of existing knowledge (Benbya et al., 2024). By deepening their core skills, students are better equipped to think creatively within their domain. As organizational research has shown, employees' skill enhancement is critical for their innovative performance (Shao et al., 2022). In GAI-enabled learning, upskilling enables students to not only generate better ideas but also to develop and refine them with the support of GAI, pushing the boundaries of their previous capabilities (Nevo et al., 2020). We therefore propose:

H8b. Upskilling is positively associated with innovative performance.

4. Research design

The increasing dependence of students on GAI represents a noteworthy emerging phenomenon, warranting a detailed exploration of its potential impacts and underlying mechanisms. Given the complexity of this issue—involving individual differences, varied GAI dependence patterns, multifaceted skill adaptations, and dual-edged performance outcomes—a single methodological approach is unlikely to provide comprehensive insights. Therefore, this study adopted a sequential mixed-methods design (Quantitative → Qualitative → Experiment) (Maier et al., 2023; Venkatesh et al., 2013). This sequential approach can combine the advantages of both quantitative and qualitative methods, addressing their respective limitations, thereby promoting a progressive investigation of the research question (Venkatesh et al., 2016).

We began with a quantitative survey (Study 1) to test our initial research model and establish generalizable insights across a large student population (Maier et al., 2023). Next, we conducted qualitative interviews (Study 2) to corroborate and complement the survey findings, with a focus on explaining unsupported hypotheses and identifying emergent contextual factors (Venkatesh et al., 2016). The insights from Study 2 informed the final phase: an experiment (Study 3) designed to test these emergent hypotheses, establish causality, and enhance the internal validity of our overall findings. This sequential design ensures triangulation and complementarity (Venkatesh et al., 2013), where each phase builds on the last to create a robust understanding of the interplay between GAI dependence, skill adaptation, and learning performance. All research procedures were approved by the authors' Institutional Review Board. Informed consent was obtained from all participants prior to their participation in each of the three studies. Participants were assured of the anonymity and confidentiality of their responses and were informed that they could withdraw from the study at any time without penalty. Fig. 3 illustrates the methodological design and phases of the sequential mixed-methods approach, and Table 2 provides a detailed overview of the three studies.

5. Study 1

5.1. Measurement development

To test the hypotheses in our initial research model, we developed a survey instrument for this study. Except for three skill adaptation scales, all constructs were measured using items adapted from previously validated studies. We slightly modified the wording of these established items to ensure their contextual relevance to GAI-enabled learning. All reflective constructs were measured with multiple items on a seven-point Likert scale (1 = "strongly disagree" to 7 = "strongly agree").

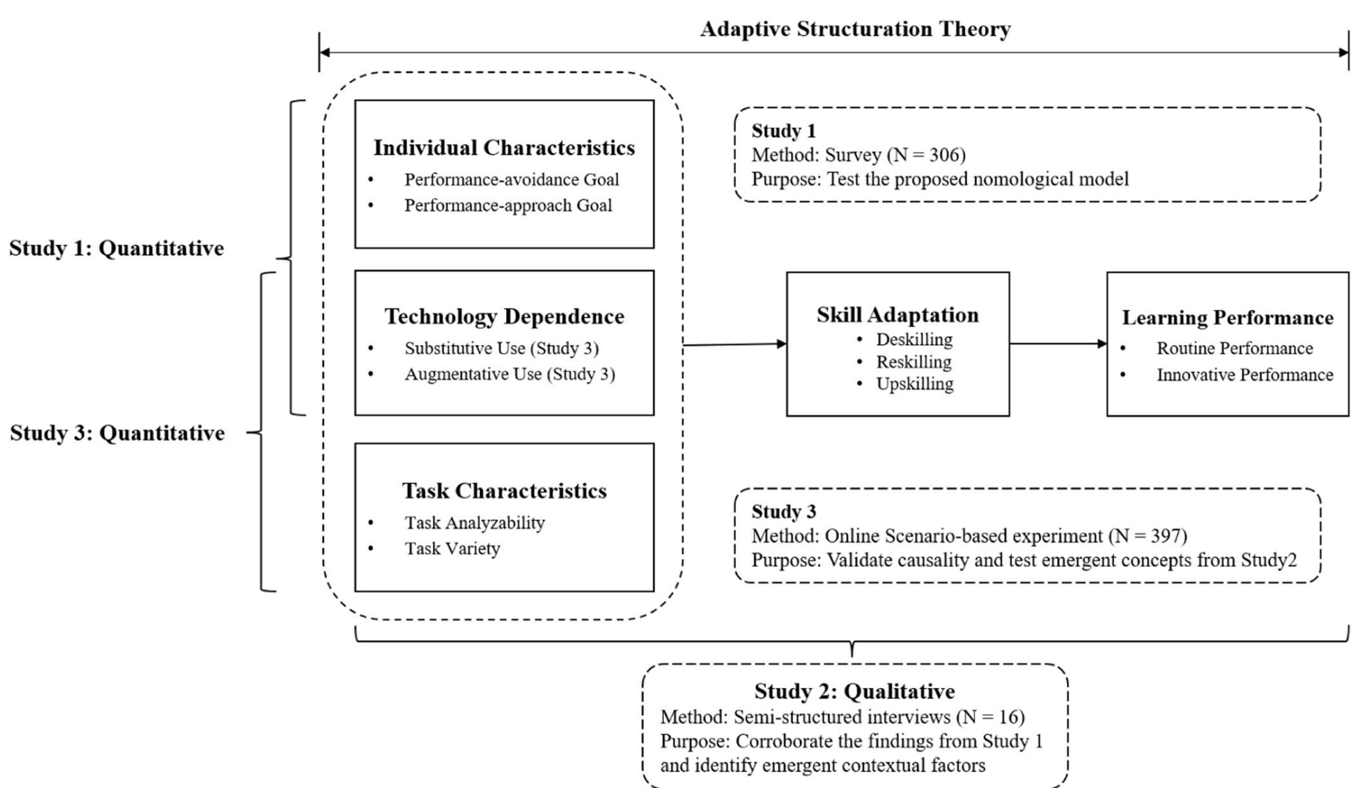

Fig. 3. Three phases of the sequential mixed-methods approach.

Table 2
Overview of the three studies.

Aspect	Study 1	Study 2	Study 3
Research Paradigm	Quantitative	Qualitative	Quantitative
Purpose	Test initial model; Establish general patterns	Corroborate and complement findings; Identify emergent factors	Provide causal evidence; Test emergent theory; Refine concepts
Methodology	Online survey	Semi-structured interviews	Online scenario-based experiment
Sample	306 GAI student users	16 GAI student users	397 GAI student users
Data Analysis	PLS-SEM	Corroboration, bracketing, coding	Regression, PLS-SEM
Contribution	Provided baseline nomological network; Identified need for explanation	Corroborate and complement findings in Study 1; Provided rationale and design basis for Study 3	Provided causal evidence for task context effect; Validated SU/AU distinction; Increased overall rigor

Specifically, we adapted five items from Shu et al. (2011) to measure technology dependence and drew from Elliot and Murayama (2008) for the performance-approach and performance-avoidance goal measures. The measure for routine performance was adapted from Ng et al. (2022), while innovative performance was measured with three items adapted from Janssen (2000). The measures of deskilling, upskilling, and reskilling were developed based on their definitions. Each measurement contains three items that reflect the impact of GAI tools on students' knowledge, skills, and abilities respectively. To ensure face validity, the initial pool of items, including the newly developed skill adaptation measures, was reviewed by five PhD candidates. These reviewers possessed expertise in both GAI technologies and survey research methodologies within the IS field. Their feedback focused critically on item clarity, theoretical relevance, wording, and potential ambiguities concerning the target population's understanding. Based on their comments, several items underwent wording refinements to enhance comprehension and precision before the instrument was finalized for data collection. All the constructs and measures are presented in Appendix A (Table A1).

5.2. Data collection and sample characteristics

The target population for Study 1 consisted of university students in China who have used GAI tools for learning tasks. This population is highly relevant as Chinese students represent a large and active user base of GAI, employing both globally prominent platforms like ChatGPT and popular domestic alternatives such as ERNIE Bot, which were the primary tools reported by our respondents (see Appendix A, Table A2). ChatGPT is an intelligent chatbot developed by OpenAI that provides timely and accurate responses to natural language questions (van Dis et al., 2023). It neared one billion monthly users by the end of 2023, indicating that ChatGPT has been prevalent globally. Similarly, ERNIE Bot, developed by Baidu, is another popular GAI tool in China. Given that ChatGPT and ERNIE Bot were two of the most widely used GAI tools in China during the period preceding Study 1's data collection, they constitute appropriate technologies for our research context.

Data collection was conducted by a reputable company (www.wjx.com). This platform was chosen for three primary reasons: (1) its extensive, pre-screened panel of university students across China, which enhances the potential representativeness of the sample to our target population; (2) its robust technical features for quality control, including IP address tracking to prevent duplicate submissions and customizable survey flow logic for screening; and (3) its ability to efficiently reach a large number of GAI users. To ensure data quality, we implemented a multi-stage screening process. The survey was administered only to participants who confirmed prior use of GAI for learning tasks. From this initial group, we excluded respondents who used GAI tools other than ChatGPT or ERNIE Bot to maintain a focused sample, as well as those who completed the survey too quickly, too slowly, or failed an attention-check question. This process yielded a final sample of 306 valid responses. The sample was composed primarily of undergraduate students (over 90 %) between the ages of 19 and 22 (83.7 %), with 71.6 % reporting ChatGPT as their primary tool and 28.4 % reporting ERNIE Bot. Notably, students majoring in engineering and technologies were the largest group, potentially reflecting the prevalence of GAI use in programming-related learning tasks. Detailed demographic statistics are available in Appendix A (Table A2).

5.3. Measurement model

We employed Partial Least Squares Structural Equation Modeling (PLS-SEM) using the SmartPLS software to analyze our data. PLS-SEM was chosen as it is well-suited for testing complex models, which aligns with our goal of understanding how skill adaptations influence performance outcomes. We assessed the measurement model's quality by examining construct reliability, convergent validity, and discriminant validity. All constructs demonstrated strong reliability, with composite reliability (CR) values exceeding the recommended threshold of 0.7 and average variance extracted (AVE) values surpassing the 0.5 threshold. Convergent validity was assessed via item loadings. One item of technology dependence (TD1), three items of routine performance (RP1, RP3, RP4), and one item of upskilling (US1) were removed due to their low loadings. After removing these items, the loadings for all the remaining items were higher than 0.6, indicating good convergent validities of the constructs. Discriminant validity was established using two criteria: first, an examination of cross-loadings showed that all items loaded more strongly on their intended construct than on any other; second, the square roots of AVE for all the constructs were greater than correlations. The data for testing the measurement model can be found in Appendix A (Tables A3 and A4). Since the data for this study were collected from single respondents at one point in time, we assessed the potential for common method bias (CMB). Procedurally, we assured respondents of anonymity to mitigate this issue. Statistically, we conducted two diagnostic tests. Harman's single-factor test showed that the first factor accounted for 20.9 % of the variance, below the 50 % threshold (Podsakoff et al., 2003). Furthermore, we applied the unmeasured latent method construct approach (Liang et al., 2007). As detailed in Appendix A (Table A5), the average substantive variance explained by the primary constructs (60.4 %) was substantially greater than the variance explained by the common method factor (0.5 %, 118:1). These results indicate that CMB is not a significant concern for this study.

5.4. Structural model

The PLS results of the structural model are summarized in Table 3. The results showed that performance-avoidance goal positively influenced deskilling ($\beta = 0.181$, $t = 3.340$), while the effect of performance-approach goal was not significant ($\beta = -0.034$, $t = 0.556$), lending support to H1a but not H1b. Instead, reskilling was significantly influenced by performance-approach goal ($\beta = 0.181$, $t = 2.717$) rather than performance-avoidance goal ($\beta = 0.010$, $t = 0.155$), thus supporting H2b rather than H2a. Upskilling was weakened by the avoidance goal and promoted by the approach goal ($\beta = -0.135$, $t = 2.837$; $\beta = 0.187$, $t = 3.203$), lending support to H3a and H3b. Additionally, technology dependence was found to be a significant positive predictor of all three skill adaptations: deskilling, reskilling, and upskilling ($\beta = 0.376$, $t = 7.663$; $\beta = 0.250$, $t = 4.353$; $\beta = 0.144$, $t = 2.863$), thus supporting H4a, H4b, and H4c. Reskilling was negatively associated with deskilling ($\beta = -0.203$, $t = 2.942$) and positively associated with upskilling ($\beta =$

Table 3
Summary of hypothesis testing results (Study 1).

Hypothesis	Path coefficients (t-value)	f^2	Supported?
H1a: PVG→DS	0.181^{***} (t = 3.340)	0.038	Yes
H1b: PPG→DS (-)	$-0.034^{n.s.}$ (t = 0.556)	0.001	No
H2a: PVG→RS (-)	$0.010^{n.s.}$ (t = 0.155)	0.000	No
H2b: PPG→RS	0.181^{***} (t = 2.717)	0.031	Yes
H3a: PVG→US (-)	-0.135^{***} (t = 2.837)	0.026	Yes
H3b: PPG→US	0.187^{***} (t = 3.203)	0.045	Yes
H4a: TD→DS	0.376^{***} (t = 7.663)	0.142	Yes
H4b: TD→RS	0.250^{***} (t = 4.353)	0.069	Yes
H4c: TD→US	0.144^{***} (t = 2.863)	0.026	Yes
H5a: RS→DS (-)	-0.203^{***} (t = 2.942)	0.043	Yes
H5b: RS→US	0.460^{***} (t = 8.720)	0.290	Yes
H6a: DS→RP	0.152^{***} (t = 3.511)	0.041	Yes
H6b: DS→IP (-)	$0.059^{n.s.}$ (t = 1.065)	0.005	No
H7a: RS→RP	0.347^{***} (t = 6.247)	0.156	Yes
H7b: RS→IP	0.259^{***} (t = 4.329)	0.073	Yes
H8a: US→RP	0.423^{***} (t = 8.157)	0.229	Yes
H8b: US→IP	0.422^{***} (t = 7.077)	0.194	Yes

Note: PPG = Performance-approach Goal, PVG = Performance-avoidance Goal, TD = Technology dependence, DS = Deskilling, RS = Reskilling, US = Upskilling, RP = Routine Performance, IP = Innovative Performance. *** $p < 0.01$. n.s. = not significant.

0.460, t = 8.720), supporting H5a and H5b, respectively. Finally, turning to performance outcomes, reskilling was positively associated with both routine performance (β = 0.347, t = 6.247) and innovative performance (β = 0.259, t = 4.329). Similarly, upskilling was positively associated with both routine performance (β = 0.423, t = 8.157) and innovative performance (β = 0.422, t = 7.077). Deskilling was only positively associated with routine performance (β = 0.152, t = 3.511), but had no significant effect on innovative performance (β = 0.059, t = 1.065). Therefore, H6a, H7a, H7b, H8a, and H8b were supported, while H6b was not supported. The model explained substantial variance in the endogenous constructs: routine performance (R^2 = 45.4 %), innovative performance (R^2 = 45.7 %), upskilling (R^2 = 36.1 %), deskilling (R^2 = 16.6 %), and reskilling (R^2 = 12.1 %).

6. Study 2

6.1. Interview design

We followed the approach recommended by Venkatesh et al. (2013) to ensure the validity of qualitative research. To achieve design validity, we employed a purposive sampling strategy with the goal of achieving maximum variation to capture a wide spectrum of GAI usage experiences (N = 16). Interested individuals were directed to a short online screening form. This form collected data to ensure diversity across several key dimensions: academic discipline (engineering, social science, arts and humanities, etc.), academic level, and self-reported GAI usage frequency (e.g., daily, weekly, monthly). This strategy was designed to ensure that the emergent themes reflected a broader and more heterogeneous range of student experiences with GAI. The design of our semi-structured interview protocol was grounded in established principles for rigorous mixed-methods research (Venkatesh et al., 2013, 2016). Consistent with their emphasis on purpose-driven design, our protocol was specifically developed to meet the objective of Study 2 within our sequential mixed-methods approach: to corroborate and complement the quantitative findings from Study 1. Furthermore, the semi-structured, open-ended nature of the interview protocol facilitated an exploratory and developmental purpose, allowing for the discovery of emergent themes beyond the initial quantitative model (Venkatesh et al., 2016). This ensured that participants could freely express their thoughts and usage experiences on the topic of GAI-enabled learning, while also providing diverse perspectives. For instance, a critical emergent insight was the significant influence of task characteristics on students' GAI dependence strategies, a finding that provided the direct empirical grounding and theoretical motivation for the design of Study 3. The semi-structured interview questions and the personal profiles of the interview participants used in this study are provided in Appendix B (Tables B1 and B2).

In the interview, most respondents stated that they were heavy users of GAI tools and had over one year of usage experience. We asked respondents about the most used GAI tools, with ChatGPT being the most frequently mentioned (N = 12), followed by GAI tools such as ERNIE Bot and Kimi, which were mentioned to a lesser extent. We noticed that the distribution of the types of GAI tools used by the respondents was consistent with the findings of Study 1. The two authors served as interviewers for approximately 45-min interviews each time. We obtained permission from the interview participants, recorded the interviews, transcribed them, and imported the transcripts into Nvivo12 for coding and analysis. Two authors independently coded the full set of transcripts. An initial multiple classification scheme was developed based on the core constructs of our research model and relevant concepts emerging from the data (Lu et al., 2024; Srivastava & Chandra, 2018). The coders then met regularly to systematically compare their assigned codes for each transcript segment, discuss interpretations, and identify any discrepancies. Following the method of Srivastava & Chandra (2018), we determined the positive or negative valence for each category and recorded any off-quadrant or divergent responses. We calculated Cohen's Kappa based on the initial, independent coding results before the discussion took place. The average Kappa value across these core constructs was 0.73, indicating substantial agreement between the coders. Appendix B (Tables B3 and B4) shows the details of our coding.

6.2. Inferences

To further validate and enrich the findings from our quantitative analysis (Study 1), we analyzed the semi-structured interview data from 16 GAI student users (Study 2). This section presents qualitative evidence that corroborates the hypotheses found significant in Study 1. After manually coding the interviewees' responses using Nvivo12, we used a bridging approach to connect the qualitative findings with the hypotheses of the quantitative research (Srivastava & Chandra, 2018).

The interviews confirmed strong student dependence on GAI tools. Moreover, participants universally agreed GAI significantly improves routine performance efficiency, though views on its impact on innovative performance were mixed. These findings and detailed evidence supporting the hypothesized relationships between technology dependence, performance goals, the three skill adaptation types (deskilling, reskilling, upskilling), and the two learning performance dimensions (routine and innovative) are summarized in Table 4. Overall, the qualitative findings provide strong corroborating evidence for the relationships proposed and supported in our quantitative model (Study 1).

6.3. Meta-inferences

We employed a bracketing approach (Lu et al., 2024; Venkatesh et al., 2013) to further analyze the interview data, seeking explanations for the hypotheses not supported in Study 1 (H1b, H2a, H6b). Moreover, following the method of Srivastava and Chandra (2018), semi-structured interviews helped us gain deeper insights into how GAI tools affect student performance. In addition to the results of Study 1, we obtained complementary insights to identify the boundary conditions under which the hypotheses of the proposed model hold true. We conducted a thorough analysis of the qualitative interview data, and the results indicate that four boundary conditions emerged in the context of how GAI tools impact skill adaptation and learning performance. For instance, viewing GAI as a tool (vs. friend) influences students' interaction style and adaptation efforts (Schweitzer et al., 2019). To present these findings concisely and clearly, we summarize the key qualitative insights and supporting evidence in Table 5. Notably, although Study 1 did not consider the impact of task characteristics, all interviewees in

Table 4
Qualitative evidence supporting constructs and hypotheses from study 1.

Construct/ Hypothesis	Summary of Qualitative Finding	Illustrative Evidence/Quote(s)
Technology dependence	Among the 16 respondents, 13 reported a heavy dependence on GAI tools, while the remaining respondents indicated varying degrees of reliance on these tools.	"When I receive a writing task, the **first thing that comes to mind** is to use ChatGPT to generate text, then I modify the text generated by ChatGPT, instead of writing it myself." (R3, female, PhD student, engineering and technologies).
Learning performance	Respondents mentioned that the use of GAI tools had a significant impact on their learning performance. In terms of routine performance, all respondents agreed that GAI significantly improved their learning efficiency.	"Sometimes, I try feeding the entire English article to GAI tools to have it generate a Chinese summary directly, which makes reading literature **highly efficient**. I don't need to read the article in detail myself to grasp the information, and the summary generated by GAI tools is much **better than my own reading**." (R16, female, undergraduate, arts and humanities).
	In terms of innovative performance, respondents had differing views; Some participants believed that GAI significantly enhanced their innovative performance. While some participants believed that although GAI improved their learning efficiency, it had no significant impact on innovative performance.	"The content generated by GAI tools sometimes gives me **inspiration** and adds **new perspectives** to my original ideas." (R8, female, undergraduate, social science and management). "GAI tools cannot provide me with good ideas. I believe good ideas still come from reading literature and communicating with peers. Although the specific steps to implement these ideas can be handled by GAI tools to **improve task efficiency**, I don't believe they offer significant help in terms of **innovation**." (R12, male, PhD student, social science and management).
	An interesting finding is that some participants were unsure about their own contribution in the process of improving performance with the help of GAI tools. This placebo effect manifested as the illusion that they easily improved their abilities with the assistance of GAI tools, making it difficult to accurately assess their own capabilities (Skulmowski, 2024).	"My concern is that I am intentionally trying to use GAI tools better, but I don't know if the quality of the content generated by GAI tools has improved due to my prompts." (R4, female, undergraduate, engineering and technologies). "During the process of using GAI tools to assist me with academic writing tasks in English, I feel that my writing ability has **indeed improved**, but I also feel that my writing ability **seems to have declined** (if there were no GAI tools). Although it has increased my **writing efficiency,** I am unsure about the extent to which my effort and ability influence the writing results generated by GAI." (R9, male, PhD student, engineering and technologies).
H1a: PVG→DS	Performance-avoidance goals are associated with greater deskilling, as users let GAI handle tasks deemed less important.	"For **those less important writing tasks**, I rely on GAI tools to help me complete them. I only need to evaluate the generated output, **without having to write it myself**." (R11, female, undergraduate, engineering and technologies).
H3a: PVG→US (-)	Performance-avoidance goals hinder upskilling, as users stop exploring once minimum requirements are met.	"Once GAI tools help me complete the task, **the course requirements are met,** and I **no longer go deeper** to explore or improve." (R13, female, undergraduate, arts and humanities).
H2b: PPG→RS	Performance-approach goals drive reskilling, as users actively learn to leverage GAI better to achieve excellent outcomes.	"With **the goal of performing better**, I **repeatedly adjust my instructions** to make the content generated by GAI tools continuously improve in order to achieve my desired outcome." (R6, female, undergraduate, social science and management).
H3b: PPG→US	Performance-approach goals foster upskilling through deeper engagement with GAI, leading to enhanced knowledge acquisition.	"**I asked GAI tools to propose a research idea**, and through this process, I **gained a deeper understanding** of research paradigms suitable for similar research contexts. In other words, with the help of GAI tools, I am becoming more familiar with this research field." (R7, female, undergraduate, social science and management).
H4a: TD→DS	GAI dependence leads to the replacement of certain user skills (deskilling), for example in information search and integration.	"GAI tools are very convenient and **have replaced much of my work**. For example, when writing papers in the past, I would search and integrate information myself, but now I directly ask GAI tools to provide me with the results. This method has somewhat reduced my need to think about the problem and **has replaced my search capabilities**." (R13, female, undergraduate, arts and humanities).
H4b: TD→RS	GAI dependence encourages users to learn new skills related to using the tool effectively (reskilling), like prompt engineering.	"During the use of GAI tools, in order to achieve the desired results, I **ask it from different angles**, **learn prompt writing**, and **consciously improve my prompt skills.**" (R2, female, undergraduate, social science and management).
H4c: TD→US	GAI dependence can lead to the enhancement of existing skills or acquisition of higher-level skills (upskilling).	"I feel that my ability in programming tasks **has improved**. I used to only program in Python, but now with the help of GAI tools, **I am able to write more difficult Java code**. Moreover, after reviewing the code generated by GAI tools, I have **deepened my understanding** and mastery of programming logic." (R10, male, undergraduate, social science and management).
H5a: RS→DS (-)	Reskilling (learning to use GAI effectively) mitigates deskilling by requiring deeper task understanding.	"**My proactive optimization of how I use GAI tools** means I need to have a more comprehensive understanding of my learning tasks. At the same time, improving the effectiveness of GAI tools

(*continued on next page*)

Table 4 (*continued*)

Construct/ Hypothesis	Summary of Qualitative Finding	Illustrative Evidence/Quote(s)
		requires me to **clarify my task ideas**, so I believe enhancing my skills in using GAI tools can **prevent a decline in my own abilities.**" (R7, female, undergraduate, social science and management).
H5b: RS→US	Reskilling facilitates upskilling, as improved GAI usage skills enhance the application of other skills (e.g., reading).	"In order to make better use of GAI tools, I work hard to **develop my ability to write prompts**. With clear prompt instructions, the output from GAI tools improves, so I use it more effectively to **enhance my literature reading skills.**" (R1, female, undergraduate, social science and management).
H6a: DS→RP	Deskilling improves routine performance by automating tasks, saving time and cognitive effort.	"With the assistance of GAI tools, my writing process has significantly accelerated. These tools have greatly saved my time and alleviated my cognitive load." (R3, female, PhD student, engineering and technologies).
H7a: RS→RP	Reskilling improves routine performance by enabling more effective and accurate task completion with GAI.	"I have **now learned to open a new chat window for each question** to avoid being influenced by the historical chat records. Additionally, I **provide some background data** so GAI tools can better learn my task context. This approach improves the quality of its responses, increasing both accuracy and relevance, helping me **complete tasks effectively**." (R14, female, undergraduate, arts and humanities).
H7b: RS→IP	Reskilling can contribute to innovative performance by sparking new ideas during exploratory use of GAI.	"**Exploratory use** of GAI tools can help me **generate some ideas** I hadn't thought of before while writing experimental reflections." (R11, female, undergraduate, engineering and technologies).
H8a: US→RP	Upskilling enhances routine performance through improved capabilities applied to tasks (e.g., better coding leading to efficiency).	"Using GAI tools to complete my programming learning tasks **improved my coding skills**, significantly **enhanced the efficiency** of completing assignments, and ultimately led to excellent grades on my submissions." (R8, female, undergraduate, social science and management).
H8b: US→IP	Upskilling supports innovative performance by refining ideas and freeing cognitive resources for creative work.	"If I have an innovative idea, GAI tools will help me **better organize my thoughts**, making the idea **more refined**." (R1, female, undergraduate, social science and management); "GAI tools help me save a considerable amount of time and effort on routine tasks, and the time and energy saved can be used for innovative tasks. Additionally, GAI tools allow me to **gain and enhance knowledge in various fields**, and based on this knowledge and skill enhancement, I will have **more innovative inspiration**." (R5, male, PhD student, life science and medicine).

Study 2 emphasized that their skill adaptation process varies for different learning tasks. Therefore, we sought to explore and analyze this important factor in depth, thereby providing a comprehensive understanding of AST in the context of GAI-enabled learning. We first recorded all the learning tasks mentioned by the respondents, and we found that these tasks could be categorized based on existing classification frameworks grounded in task-technology fit theory (Goodhue & Thompson, 1995). After a detailed analysis of the learning tasks we recorded, two coders ultimately decided to use a 2 × 2 framework with task variety and task analyzability (Crumbly et al., 2025; Daft & Macintosh, 1981). We provided this classification framework in Fig. 4, along with representative examples of the task names mentioned by respondents in the qualitative interviews, placed in the four quadrants. This finding regarding the impact of task characteristics served a crucial developmental purpose within our overall mixed-methods program (Venkatesh et al., 2013, 2016), directly informing the conceptualization and design of Study 3, which experimentally manipulates task types (based on variety and analyzability) to test these emergent insights.

7. Study 3

7.1. Research model and hypothesis development

This study employs a sequential mixed-methods design to comprehensively investigate students' skill adaptation to GAI. While Study 1 established the general significance of technology dependence in influencing skill adaptation pathways (deskilling, reskilling, upskilling), it treated dependence as a relatively overall construct.

A key insight emerging from the subsequent qualitative phase (Study 2) was that students mentioned task characteristics shaped the way they depend on and utilize GAI tools (see Section 6.3, Table 5, Fig. 4). Participants described shifting their interaction styles—sometimes relying heavily on GAI to automate tasks, other times using it more interactively to enhance their own efforts—based on whether the task was perceived as analyzable or varied. Revealing that a single measure of technology dependence could obscure differential impacts on skill adaptation, this empirical finding necessitated a clear distinction between ***passive*** dependence and ***strategic*** utilization. Therefore, Study 3 was designed with a specific developmental purpose integral to our mixed-methods approach (Venkatesh et al., 2013, 2016): to experimentally investigate how task characteristics causally influence the adoption of distinct GAI use patterns, and consequently, their differential impacts on skill adaptation. The research model of Study 3 is presented in Fig. 5.

To operationalize the insights from Study 2, we draw upon the established distinction between AI for automation vs. augmentation to conceptualize these GAI dependence patterns (Leyer & Schneider, 2021; Raisch & Krakowski, 2021; Rosing & Zacher, 2017). This framework posits that AI can be used to **automate** human work, reflecting a substitution of labor to improve efficiency, or to **augment** human work, reflecting a synergy between human expertise and AI capabilities to enhance performance (Raisch & Krakowski, 2021). Guided by this, we define two distinct use patterns for Study 3. Specifically, **substitutive use (SU)** reflects a passive, automation-focused dependence where GAI is used to replace students' effort and perform significant parts or the entirety of a task (Guo et al., 2024). In contrast, **augmentative use (AU)** represents a strategic, synergistic engagement where GAI is used to enhance or supplement the students' own cognitive processes and capabilities (Guo et al., 2024). While both SU and AU are forms of

Table 5
Qualitative insights on unsupported hypotheses and boundary conditions.

Unsupported Hypothesis	Qualitative Insight/ Explanation	Illustrative Quote(s)
H1b: PPG → DS	The relationship between PPG and deskilling may be moderated by the **temporal orientation of goals**. Students with performance-approach goals might accept deskilling for short-term tasks if GAI ensures excellent results efficiently.	“My learning goals require me to achieve a certain level and attain excellent results. If using GAI tools can help me achieve the desired level **without requiring me to have very high personal abilities**, I will adjust to be goal-oriented. **As long as the result is good**, I am happy to let GAI tools **replace me**.” (R7, female, undergraduate, social science and management). “At my current stage, some translation or **reading abilities are just an intermediate process** to achieve my final goal, **not my ultimate goal**. Therefore, it doesn’t matter to me if these abilities are **replaced** by GAI tools at this stage.” (R12, male, PhD student, social science and management).
H2a: PVG → RS	Interviews indicated that even avoidance-motivated students might engage in reskilling due to **expected benefits** like task completion speed, even if not aiming for excellence.	“Although my learning goals are not set very high, and I think **completing the task is enough**, I still want to **refine my level of using GAI tools** to generate content that aligns with my ideas, rather than just using whatever is randomly generated.” (R11, female, undergraduate, engineering and technologies).
H6b: DS → IP	Deskilling’s potential negative impact on innovative performance seems counteracted by: (1) **Freeing cognitive resources** for innovation, and (2) GAI **expanding knowledge domains**, sparking inspiration.	“When completing creative tasks, I need GAI tools to replace me in handling some routine tasks, such as organizing language, to **release my cognitive load**. I don’t need to spend so much energy on those routine tasks, thus freeing up more energy and focus for **exploring more creative tasks**.” (R1, female, undergraduate, social science and management). “GAI tools are trained on extensive corpora, so the texts generated by GAI tools contain professional vocabulary or fields that I have not encountered before. They can cross professional barriers and **teach me content from other fields** relevant to my tasks, giving me a **broader perspective and more inspiration**. Compared to my own writing, the content generated by GAI tools may be more creative.” (R6, female, undergraduate, social science and management).

Table 5 (*continued*)

Boundary Condition	**Observed Adaptation Pattern**	**Illustrative Evidence/ Quote(s)**
Task Characteristics -Analyzable/ Low Variety	Tendency to automate tasks using GAI. Abilities are largely **replaced (Deskilling)**. Efficiency improves, but concerns about skill erosion (e.g., expressing opinions, organizing academic language) exist, though sometimes disregarded if GAI is effective.	*Example Tasks:* Translating texts, Editing texts, Writing lab reports “In English writing and editing, I feel that many of my skills have been **replaced by GAI tools**, which has greatly **improved efficiency**; however, after using them for a long time, I sometimes feel that I am **unable to effectively express my opinions**, in other words, I cannot organize academic language in my papers well. However, when GAI tools are able to replace me effectively in these tasks, **I no longer mind this impact**.” (R9, male, PhD student, engineering and technologies).
Task Characteristics -Analyzable/ High Variety	Active engagement with GAI. Users explore ways to leverage tools, leading to both **Reskilling** (e.g., providing feedback, breaking down tasks, optimizing prompts) and **Upskilling** (e.g., improved programming ability, reduced need to memorize functions).	*Example Tasks:* Programming, Language learning “In programming tasks, I **continuously provide feedback** to GAI tools after each debugging session until the code works. I also **consciously break down tasks**, **set up the top-level framework**, and then let GAI tools assist me in completing the task **in stages.** With GAI tools, I delegate some specific coding tasks to them, which **reduces many functions I need to memorize**. After using it more, I found that I could directly type these functions, so ultimately my **programming ability has improved**.” (R12, male, PhD student, social science and management).
Task Characteristics -Unanalyzable /Low Variety	Students may no longer need to generate novel ideas themselves, relying on GAI’s creativity. This would lead to improved efficiency but also **Deskilling** of independent, deep thinking and innovation skills.	*Example Tasks:* Creative design, Generating ideas for specific problems “When I need to extract ideas from existing materials, **I directly ask GAI** tools to help me summarize and refine them, which improves my **efficiency**. I no longer think independently, search for information, or meticulously devise solutions, nor do I spend as much time thinking about how to innovate. Gradually, my **ability for deep thinking has decreased.**” (R6, female, undergraduate, social science and management).
Task Characteristics -Unanalyzable /High Variety	Strategic use of GAI to stimulate creativity and integrate knowledge. Involves **Reskilling** (e.g.,	*Example Tasks:* Conducting systematic scientific research, Interdisciplinary innovation, Software

(*continued on next page*)

Table 5 (*continued*)

	designing precise, optimized prompts, understanding GAI logic) and **Upskilling** (e.g., learning ideas/methods from other fields, gaining inspiration).	architecture development "There are many subfields in our biology discipline. For example, those working on protein structure research may not understand how teams conducting mouse experiments carry out their work and experiments. When faced with interdisciplinary challenges like these, I use methods such as **precise and continually optimized prompts** to inquire deeply with GAI tools about which methods from other fields can help solve my problem. Through this process, I have **learned many ideas for designing experiments**, which has provided me with a lot of **inspiration**. In this process, I also explored the **underlying operational logic of GAI tools** and learned **how to design prompts**." (R5, male, PhD student, life science and medicine).
GAI Performance	81 % of respondents emphasized that the performance of GAI tools has a significant impact on their usage and performance. Perceived capabilities and reliability of specific GAI tools significantly affect usage patterns, adaptation, and task-tool matching.	"Before the release of GPT, I actually had already learned about and used related GAI tools, such as the early version of GPT3. I found that it couldn't handle relatively complex tasks, like mathematical problems at all. Since I work in software development and architecture design, I have high requirements for the logical capabilities of GAI tools. I found that Llama has much stronger logical abilities compared to models like ChatGPT, and **it performs better in completing logical tasks**. Therefore, I **engage in more exploration** during interactions with it, which makes my usage **more diverse.**" (R15, male, undergraduate, social science and management). "I often **use different GAI tools based on different tasks**, and I think this approach **works better**. For example, I think ChatGPT is better suited for writing code, while for language learning and conversations, I find Doubao works better. For some more everyday questions, I would ask ERNIE Bot." (R13, female, undergraduate, arts and humanities).
Anthropo-morphism	Viewing GAI as a tool vs. a partner/friend influences interaction style and adaptation efforts. Most students (N = 12) treat GAI tools as tools and interact with them in a more mechanical manner. While a few participants, however, feel that GAI tools are more like collaborative partners (N = 3) or even friends (N = 1). Partner-view associated with more conscious reskilling.	"I just treat ChatGPT as a **tool**. We use it in a very **direct and mechanical** way. For example, I usually say 'help me summarize this material' or 'help me generate something' like that." (R14, female, undergraduate, arts and humanities). "I feel that ChatGPT is my **partner** in learning. I recognize and expect certain abilities from it. I **consciously verify its output** during interactions and **write my own prompts.** I feel that my ability to use ChatGPT has **improved**." (R1, female, undergraduate, social science and management). "ChatGPT is my friend. I **speak politely** to it. When it gives me wrong answers, I **patiently and politely** engage with it **step by step** until it provides the correct answer, even though I know this won't affect its output." (R16, female, undergraduate, arts and humanities).
Concerns about GAI tools' ability to replace and degrade themselves:	Anxiety about skill degradation influences adaptation. Concerned students may limit GAI use or practice skills independently; unconcerned students may accept deskilling and focus on GAI-specific skills.	"I feel that GAI tools make everyone's expression **more uniform and homogenized.** I am **worried** that in terms of writing, my ability to express myself and my uniqueness, as well as my writing skills, will decline. So, sometimes I may **intentionally reduce its involvement** in text creation and **try to write something myself** to complete some tasks." (R14, female, undergraduate, arts and humanities). "As long as the use of GAI tools results in high-quality outcomes, I am open to having my writing skills supplemented or replaced by these tools. To achieve my objectives and obtain better responses, I am willing to approach the task from various perspectives and invest in learning diverse methods to optimize my instructions." (R2, female, undergraduate, social science and management).

dependence, we theorize that their underlying mechanisms lead to different consequences for skill development.

SU refers to the automation of task processes (Guo et al., 2024), where students delegate the entirety of problem-solving and task execution to GAI systems. In such cases, GAI replaces the need for students to apply their own skills. This over-reliance on automation can lead to the gradual decline of essential skills due to insufficient use (Rinta-Kahila et al., 2023; Skaria et al., 2020). Ultimately, technology replaces human capability, leading to a substitution effect. Therefore, we propose the following hypothesis:

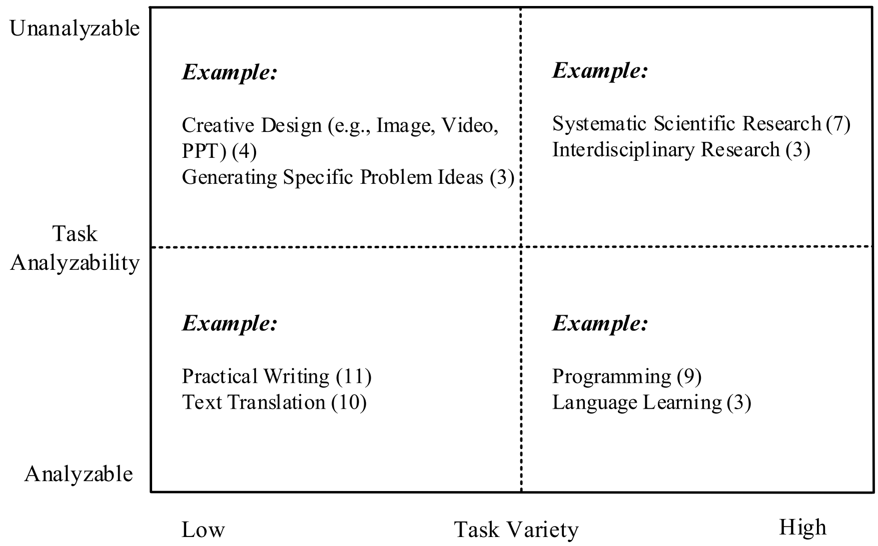

Fig. 4. A variety/analyzability framework for learning tasks assisted by GAI tools. **Note**: The numbers in brackets indicate the number of times the respondents mentioned the learning task during the interview.

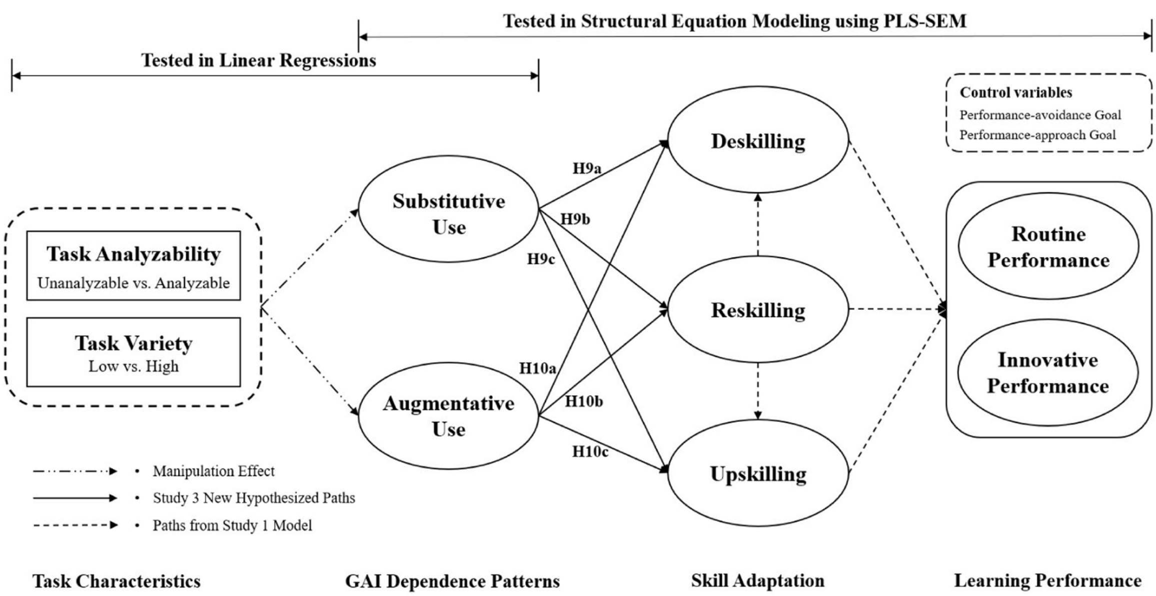

Fig. 5. Research model of Study 3.

H9a. Substitutive use is positively associated with deskilling.

Furthermore, SU undermines the motivation to acquire new skills. Specifically, SU reflects a form of passive technology dependence, where students bypass cognitively demanding learning activities such as knowledge extraction and reflection processes (Guo et al., 2024). When the GAI tool assumes full responsibility for the task, there is little incentive for the student to learn new competencies (e.g., advanced prompting, output verification) to use the tool more effectively. This lack of active engagement inhibits the exploratory behavior necessary for reskilling (Leon, 2023). We therefore propose:

H9b. Substitutive use is negatively associated with reskilling.

Similarly, the impact of SU on existing skills is detrimental. Upskilling requires that existing competencies are actively practiced. However, SU removes the opportunity for such engagement. By handing over tasks to GAI, students are not applying their domain knowledge in ways that would improve their abilities (Fügener et al., 2021). Prior research indicates that when AI is used for full automation, it can have a detrimental effect on skill enhancement and self-determined motivation (Passalacqua et al., 2025). Accordingly, we propose:

H9c. Substitutive use is negatively associated with upskilling.

Conversely, AU reflects the augmentation role of GAI, whereby the tool is employed to enhance, rather than substitute, students' own capabilities (Guo et al., 2024). This usage pattern aligns with the concept of augmented intelligence, where the technology complements and extends human intellect rather than replacing it (Raisch & Krakowski, 2021). When students use GAI augmentatively, they remain the primary agent in the learning process, using the tool to scaffold their own thinking (French & Shim, 2024). By maintaining human agency, AU should protect against the skill substitution effect associated with deskilling. Thus, we hypothesize:

H10a. Augmentative use is negatively associated with deskilling.

Moreover, to use GAI as an augmentation tool, students must learn to guide the technology, critically assess its outputs, and integrate it into their workflow (Guo et al., 2024). AU is an active appropriation of technology, where the user must learn to interact with the system in an exploratory way (Leyer & Schneider, 2021). This motivates users to acquire new skills, such as how to decompose complex problems for the GAI, design effective prompts, and critically evaluate GAI outputs. An intentional effort to develop these new competencies is the core of reskilling. Therefore, we hypothesize:

H10b. Augmentative use is positively associated with reskilling.

AU also creates a powerful augmentative effect that enhances a student's original capabilities. Specifically, AU restructures the task process to support human agency and creativity (Raisch & Krakowski, 2021). By functioning as a cognitive scaffold, GAI can enable students to allocate more of their finite cognitive resources to higher-order aspects of their learning (Guo et al., 2024). Such a strategic partnership empowers students to tackle more challenging problems, thereby deepening their understanding within their core domain (Passalacqua et al., 2025). Therefore, we propose:

H10c. Augmentative use is positively associated with upskilling.

7.2. Methodology

We employed an online experiment to test the new hypotheses proposed in Study 3 and the prior hypotheses in Study 1, building upon the insights from Study 2. The experiment was specifically designed to induce different types of technology dependence patterns (AU and SU) by manipulating task characteristics. This experimental intervention allows for a rigorous examination of the causal relationships between constructs (Rob et al., 2025). Guided by the findings from Study 2, which indicated that task characteristics influence how students adapt their GAI use, we adopted a 2 (task analyzability: unanalyzable vs. analyzable) × 2 (task variety: low vs. high) between-subjects factorial design. Our manipulation aimed to elicit greater AU under conditions of unanalyzable tasks and high variety, while eliciting greater SU under conditions of analyzable tasks and low variety. This experimental inducement enables us to test the differential causal pathways associated with these two distinct forms of GAI dependence. Specifically, we anticipated that exposure to unanalyzable and high-variety tasks would lead to higher reported levels of AU, whereas exposure to analyzable and low-variety tasks would result in higher reported levels of SU. Furthermore, we expected that SU would, in turn, positively influence deskilling while negatively influencing reskilling and upskilling. Conversely, we anticipated that AU would negatively influence deskilling but positively influence both reskilling and upskilling. This experimental design directly addresses the developmental insights from Study 2, testing the emergent theory regarding task-contingent GAI dependence patterns and their skill adaptations (Venkatesh et al., 2013, 2016). Study 3 comprised two phases: a pre-test phase to validate the manipulations and a main online experiment phase to test the hypotheses.

7.2.1. Experimental stimuli

The experiment was situated in the GAI-enabled learning context, consistent with Studies 1 and 2. Four scenarios depicting different learning tasks were created to operationalize the 2 × 2 design, manipulating task analyzability and task variety based on established task taxonomies (Crumbly et al., 2025; Daft & Macintosh, 1981) (Detailed operationalization of scenarios in Appendix C, Table C1). To manipulate task analyzability, we distinguished between **analyzable** tasks (clear procedures, objective solutions, e.g., summarizing factual text) and **unanalyzable** tasks (ambiguous, requiring judgment/intuition, e.g., generating novel creative concepts). To manipulate task variety, we distinguished between **low variety** tasks (repetitive, predictable workflow, e.g., formatting citations according to a style guide) and **high variety** tasks (novel events, diverse problems, e.g., developing a complex software architecture with evolving requirements).

A pre-test involving 120 participants, randomly assigned to one of the four scenarios, was conducted. Participants rated the perceived analyzability and variety of the task described in their scenario (7-point Likert scales). A *t*-test analysis (two-tailed) indicated significant differences in the mean score for task analyzability between the unanalyzable tasks ($M_{unanalyzable} = 3.53$, $SD = 1.41$) and the analyzable tasks ($M_{analyzable} = 5.14$, $SD = 0.91$, $t(118) = -7.437$, $p < 0.001$), as well as between the task variety scores for the low variety tasks ($M_{low\ variety} = 5.15$, $SD = 1.12$) and the high variety tasks ($M_{high\ variety} = 5.71$, $SD = 0.53$, $t(118) = -3.503$, $p < 0.001$). These significant differences, coupled with large effect sizes in the intended directions, confirmed the validity of our manipulations.

7.2.2. Procedure and sample

Following the pre-test and refinement of materials, the main online experiment was conducted through an MTurk-like platform (Credamo) in China. This platform facilitated access to the target population of Chinese university students, consistent with the rationale outlined for Study 1. The procedure, detailed in Appendix C (Fig. C1), involved the following steps: (1) Introduction: Participants were informed about the study's purpose and duration, and provided consent. (2) Random Assignment: Participants were randomly allocated to one of the four experimental conditions. Then they read the specific scenario describing the learning task corresponding to their assigned condition. (3) Manipulation Check: Participants answered questions assessing perceived task analyzability and variety. (4) Measurement: Participants completed questionnaires measuring the other constructs and demographics.

To ensure alignment with the study's focus on GAI tools usage, the first page of the questionnaire included a screening question: "Have you ever used generative AI tools (e.g., ChatGPT, ERNIE Bot, DeepSeek, Doubao) to complete learning tasks?" Only respondents who answered "Yes" were permitted to proceed. Following quality checks—excluding incomplete responses, outliers with short/long completion times, and failures in attention-filter questions—a final sample of 397 valid responses was retained for analysis. Demographic characteristics are summarized in Appendix C (Table C3). The majority of participants reported DeepSeek (45.6 %) as their primary GAI tool, followed by Doubao (25.7 %) and ChatGPT (11.8 %). Notably, respondents aged 21–23 years dominated the sample (51.4 %), with 91.5 % being undergraduates and 40.6 % majoring in social sciences and management, reflecting GAI's growing adoption in interdisciplinary learning contexts. Additionally, 46.3 % of participants had 1–2 years of experience with GAI tools, and 41.8 % reported monthly or more frequent usage, underscoring the integration of these technologies into routine academic workflows.

7.2.3. Measures

All constructs were reflective and measured using multiple items on 7-point Likert scales (1 = strongly disagree, 7 = strongly agree). Differing from Study 1, the operationalization of technology dependence was refined in Study 3. Measures for other core constructs were kept consistent with Study 1 (see Appendix A, Table A1). The key dependent variables for the elicitation check and independent variables for the main hypothesis tests were SU and AU. These were measured using scales adapted from Guo et al. (2024) to fit the GAI-enabled learning context. The specific items comprising these measures are presented in Appendix C (Table C2).

7.3. Results

Data analysis proceeded in stages: first, confirming the manipulations and their intended effect on eliciting GAI dependence patterns, and second, testing the hypotheses using PLS-SEM.

7.3.1. Manipulation check

First, a *t*-test analysis (two-tailed) indicated significant differences in the mean score for task analyzability between the unanalyzable tasks ($M_{unanalyzable} = 3.54$, SD = 1.39) and the analyzable tasks ($M_{analyzable} = 5.48$, SD = 0.89, t(395) = -16.530, $p < 0.001$), as well as between the task variety scores for the low variety tasks ($M_{low\ variety} = 4.70$, SD = 1.10) and the high variety tasks ($M_{high\ variety} = 5.69$, SD = 0.53, t(395) = -11.356, $p < 0.001$). The mean differences in the expected directions and the large associated effect sizes confirm the validity of our experimental manipulation.

We then examined whether the task manipulations successfully elicited the intended GAI use patterns (SU and AU). Specifically, we expected that participants assigned to the analyzable and low-variety task conditions would report higher levels of SU, while those assigned to the unanalyzable and high-variety task conditions would report higher levels of AU. As shown in Table 6, our regression analyses confirm that the task manipulations significantly influenced GAI use patterns. For each dependent variable, we tested the effect of the task analyzability treatment (dummy coded: 0 = unanalyzable, 1 = analyzable) and the task variety treatment (dummy coded: 0 = low variety, 1 = high variety). Two models were run for each treatment effect: one with the treatment variable only (Models 1a, 2a, 3a, 4a) and one including control variables (age, usage frequency, and learning advantage) (Models 1b, 2b, 3b, 4b). Collectively, these findings support the effectiveness of our experimental design in eliciting the intended GAI use patterns (SU vs. AU) based on the manipulated task characteristics. This provides a solid foundation for testing the subsequent hypotheses regarding the differential effects of SU and AU on skill adaptation in the structural model. Notably, the manipulation of task characteristics, on its own, explained a relatively small portion of the variance in SU and AU. While this confirms a statistically significant causal link, the modest effect size suggests that task context is one of several important factors influencing how students use GAI. As our qualitative findings indicated, other factors such as the perceived performance of the GAI tool and individual differences in technology perceptions (e.g., viewing GAI as a tool vs. a partner) likely play a key role in shaping GAI usage patterns in real-world settings.

7.3.2. Measurement model

Similar to Study 1, we used PLS to test the measurement and structural model. We checked the CR and AVE for each construct to assess their reliabilities, and all of them exceeded the suggested threshold values of 0.7 and 0.5, respectively. Convergent and discriminant validities were assessed by examining whether the item loadings on their respective constructs were sufficiently high and higher than the loadings on other constructs. Three items of routine performance (RP1, RP3, RP4), two items of augmentative use (AU4, AU5), and two items of substitutive use (SU5, SU6) were removed due to their low loadings. After removing these items, the loadings for all the remaining items were higher than 0.6, indicating good convergent validities of the constructs. Further, the item loadings on their respective constructs were higher than the loadings on other constructs, and the square roots of AVE for all the constructs were greater than correlations, suggesting good discriminant validities of the constructs. The validation data for the measurement model can be found in Appendix D (Tables D1 and D2). Similarly, we checked for CMB. Harman's single-factor test indicated that the first factor explained 26.4 % of the total variance, well under the 50 % threshold (Podsakoff et al., 2003). The unmeasured latent method construct test also confirmed that the average substantive variance (69.7 %) was significantly higher than the average method variance (0.9 %), with a ratio of approximately 74:1 (see Appendix D, Table D4 for details) (Liang et al., 2007). Therefore, CMB is unlikely to have confounded the findings of this study.

Table 6
Results comparing manipulations of substitutive use and augmentative use.

Variables	DV: Substitutive use		DV: Augmentative use	
Task analyzability	Model (1a)	Model (1b)	Model (2a)	Model (2b)
Treatment:	0.278***	0.255***	− 0.152*	− 0.134**
0: unanalyzable	(0.100)	(0.097)	(0.075)	(0.071)
1: analyzable				
Control variables	No	Yes	No	Yes
R^2	0.019	0.085	0.01	0.103
Variables	DV: Substitutive use		DV: Augmentative use	
Task variety	Model (3a)	Model (3b)	Model (4a)	Model (4b)
Treatment:	− 0.174*	− 0.194**	0.152**	0.155**
0: low variety	(0.101)	(0.098)	(0.075)	(0.072)
1: high variety				
Control variables	No	Yes	No	Yes
R^2	0.008	0.078	0.01	0.106

Note: *p < 0.1, ** p < 0.05, *** p < 0.01.

7.3.3. Structural model

PLS results of the structural model are summarized in Table 7. First, we examined the hypothesized effects of GAI use patterns. The results showed that substitutive use significantly increased deskilling (β = 0.208, t = 3.702) and reduced both reskilling (β = − 0.107, t = 1.974) and upskilling (β = − 0.138, t = 3.245), supporting H9a, H9b, and H9c. While augmentative use showed no significant impact on deskilling (β = − 0.091, t = 1.374), augmentative use strongly enhanced reskilling (β = 0.444, t = 7.539) and upskilling (β = 0.352, t = 6.659), supporting H10b and H10c but rejecting H10a. Next, we examined the relationships from our initial model. Reskilling reduced deskilling (β = − 0.104, t = 1.794) and promoted upskilling (β = 0.370, t = 7.564), validating H5a and H5b. Instead, deskilling showed no significant effect on routine performance (β = 0.025, t = 0.595) and innovative performance (β = − 0.013, t = 0.231), rejecting H6a and H6b. Finally, reskilling and upskilling both enhanced innovative performance (β = 0.235, t = 3.790; β = 0.380, t = 4.329) and routine performance (β = 0.262, t = 4.481; β = 0.449, t = 6.094), supporting H7a, H7b, H8a, and H8b. All the variables jointly explain 40.3 % of the variance in routine performance. The model explains 52.1 % of the variance in upskilling, 40.3 % in routine performance, 31.2 % in innovative performance, 27.6 % in reskilling, and 10.7 % in deskilling.

8. Discussion and conclusion

This study used a sequential mixed-methods approach (Quantitative Survey → Qualitative Interviews → Quantitative Experiment) to investigate the relationship between students' learning goals, GAI dependence, skill adaptation processes, and subsequent learning performance.

Table 7
Summary of hypothesis testing results (study 3).

Hypothesis	Path coefficients (t-value)	f^2	Supported?
H9a: SU→DS	0.208*** (t = 3.702)	0.036	Yes
H9b: SU→RS (-)	− 0.107** (t = 1.974)	0.012	Yes
H9c: SU→US (-)	− 0.138*** (t = 3.245)	0.029	Yes
H10a: AU→DS (-)	− 0.091 n.s. (t = 1.374)	0.006	No
H10b: AU→RS	0.444*** (t = 7.539)	0.194	Yes
H10c: AU→US	0.352*** (t = 6.659)	0.154	Yes
H5a: RS→DS (-)	− 0.104* (t = 1.794)	0.009	Yes
H5b: RS→US	0.370*** (t = 7.564)	0.207	Yes
H6a: DS→RP	0.025 n.s. (t = 0.595)	0.001	No
H6b: DS→IP (-)	− 0.013 n.s. (t = 0.231)	0.000	No
H7a: RS→RP	0.262*** (t = 4.481)	0.073	Yes
H7b: RS→IP	0.235*** (t = 3.790)	0.051	Yes
H8a: US→RP	0.449*** (t = 6.094)	0.202	Yes
H8b: US→IP	0.380*** (t = 4.329)	0.125	Yes

Note: SU = Substitutive Use, AU = Augmentative Use, DS = Deskilling, RS = Reskilling, US = Upskilling, PVG = Performance-avoidance Goal, PPG = Performance-approach Goal, RP = Routine Performance, IP = Innovative Performance. *p < 0.1, ** p < 0.05, *** p < 0.01, n.s. = not significant.

Grounded in AST and enriched by perspectives from LPT, AGT, and the automation-augmentation framework, our research progressively provides a comprehensive understanding of how students adapt to GAI, the factors influencing these adaptations, and their performance consequences.

8.1. Key findings

The first objective of this study was to identify key skill adaptation strategies (deskilling, reskilling, upskilling) and explore how individual characteristics (performance-approach and -avoidance goals) and overall technology dependence influence these adaptations. Study 1 provided the initial test of these relationships within our nomological network (Fig. 2). Consistent with AGT and prior adaptation research, students' learning goals significantly shaped their adaptation patterns (Leung et al., 2022; Nevo et al., 2021). Performance-avoidance goals positively predicted deskilling (H1a supported) and negatively predicted upskilling (H3a supported), suggesting a tendency towards effort minimization and avoidance of skill enhancement when motivated by fear of failure. Conversely, performance-approach goals positively predicted reskilling (H2b supported) and upskilling (H3b supported), indicating that students aiming to outperform others are more likely to actively acquire new GAI-related skills and enhance existing ones. General technology dependence on GAI tools exhibited a multifaceted influence, positively predicting deskilling (H4a supported), reskilling (H4b supported), and upskilling (H4c supported). This suggests that GAI dependence drives adaptation, including both the potential erosion of existing skills and the impetus to learn new ways of working with the technology and to enhance existing competencies through its use. Moreover, the results supported the hypothesized interrelationships between adaptation dimensions. Reskilling was negatively associated with deskilling (H5a supported) and strongly positively associated with upskilling (H5b supported) in both Study 1 and Study 3. This aligns with AST's premise that newly appropriated structures (reskilling) can modify existing ones (deskilling, upskilling) (Schmitz et al., 2016), suggesting that actively learning to leverage GAI can counteract skill loss and boost existing skill enhancement. Notably, the hypothesized negative link between performance-approach goals and deskilling (H1b) was not supported in Study 1. This may be because students with performance-approach goals are also affected by the skill erosion caused by using GAI tools (Rinta-Kahila et al., 2023), although their higher self-efficacy compared to students with performance-avoidance goals might offset some of the deskilling caused by GAI (Huang, 2016). Our qualitative data from Study 2 corroborates this finding; even students with performance-approach goals might strategically accept deskilling for non-central tasks if doing so guarantees the high-quality outcomes necessary for their ultimate achievements. Similarly, the hypothesized negative link between performance-avoidance goals and reskilling (H2a) was also not supported. An explanation could be that students with performance-avoidance goals tend to view using GAI tools as a means to avoid failure (Van Laar et al., 2019). In this case, they may develop new skills to achieve the goal of avoiding failure, thus not exhibiting a clear negative association with reskilling. Interview data also suggests that even avoidance-motivated students may engage in some reskilling. They might learn to better use GAI tools not to excel, but as a coping mechanism to ensure their work meets the minimum requirements for task completion, thus avoiding failure.

Second, our objective involved differentiating routine and innovative performance and assessing the impact of the three skill adaptations, with findings drawn from both Study 1 and Study 3. Both reskilling and upskilling consistently emerged as significant positive antecedents of both routine performance (H7a, H8a supported) and innovative performance (H7b, H8b supported) across both Study 1 and Study 3. This robust finding indicates the importance of skill augmentation and reconfiguration for leveraging GAI effectively, contributing positively to both efficiency-oriented and creativity-oriented learning outcomes. This contrasts with previous studies that focused solely on the effect of GAI usage on routine performance (Abbas et al., 2024; Rehman et al., 2024; Shahzad et al., 2024). However, the consequences of deskilling were ambiguous. The hypothesized negative impact of deskilling on innovative performance (H6b) was not supported in either Study 1 or Study 3. While deskilling might intuitively seem detrimental to innovation, counteracting mechanisms likely exist. Our qualitative findings suggest potential counteracting mechanisms. For instance, GAI interaction was reported to expose students to novel information and cross-disciplinary concepts, which could spark inspiration and thereby offset the direct negative effects of deskilling on innovation. Moreover, the link between deskilling and routine performance (H6a) yielded inconsistent results. It was positive and significant in Study 1, supporting the theoretical argument that task simplification via skill substitution enhances efficiency and aligning with qualitative findings. However, this relationship was non-significant in Study 3's experiment. This discrepancy may stem from key methodological differences between the two studies. Study 1 captured students' general perceptions based on self-reported, aggregated experiences with a wide array of routine tasks over time, where the efficiency benefits of delegating work (deskilling) are easily recalled. In contrast, Study 3's controlled experimental setting focused on a specific task scenario, which may not have fully captured the context in which the positive link between deskilling and routine performance becomes salient.

Finally, our third objective, strongly informed by Study 2, focused on how task characteristics shape distinct GAI dependence patterns (SU and AU) and their subsequent effects on skill adaptation. Interviews revealed that students adapt their GAI use based on task context, often employing GAI substitutively for analyzable/low-variety tasks and augmentatively for unanalyzable/high-variety ones. This motivated the distinction between SU and AU, drawing on the automation-augmentation framework, and the design of Study 3. Furthermore, Study 3 confirmed these relationships experimentally. Task manipulations significantly influenced the adoption of SU vs. AU as intended. These use patterns also differentially impacted skill adaptation: SU significantly increased deskilling (H9a supported) while decreasing reskilling (H9b supported) and upskilling (H9c supported), supporting the view that automation-focused dependence can erode skills. Conversely, AU strongly promoted reskilling (H10b supported) and upskilling (H10c supported), highlighting the skill-enhancing potential of augmentative GAI use. However, the expected negative relationship between AU and deskilling (H10a) was not supported. While AU promotes positive skill development, it did not significantly reduce perceived deskilling in our experiment. This could be because the strong positive effects of AU on reskilling might indirectly mitigate deskilling (via the supported H5a path), making the direct negative path statistically non-significant.

Collectively, these findings across our three studies provide a detailed account of skill adaptation in the context of GAI dependence, highlighting the interplay of individual, technological, and contextual factors, and offering explanations for both expected and unexpected relationships within the proposed nomological network. Thus, our study provides a mechanism-based understanding crucial for managing information behaviors in the age of GAI.

8.2. Theoretical implications

This study yields several significant theoretical contributions relevant to IS and related fields by examining student adaptation to GAI through a sequential mixed-methods lens. First, it contributes to the literature on human adaptation to intelligent technologies by conceptualizing, operationalizing, and empirically validating individual skill adaptation resulting from GAI dependence. Previous studies have typically adopted a technology-determinism perspective, assuming that students' dependence on GAI directly affects positive and negative outcomes (Larson et al., 2024; Skulmowski, 2024), neglecting the perspective of user adaptation and coping mechanisms. Moreover, while prior research acknowledges technology's impact on skills (e.g., Dornelles et al., 2023; Li, 2022), it tends to either neglect individual-level investigations or simplify the outcome to a dichotomy between skill enhancement and decline (Lee et al., 2024; Naamati-Schneider & Alt, 2024; Wu et al., 2024). In contrast, grounded in LPT, this study contributes to the literature by conceptualizing and operationalizing three distinct individual-level skill adaptation processes relevant to GAI use: deskilling, reskilling, and upskilling. Our multi-phase study provides empirical validation for these constructs and their measures, offering a reliable instrument for future research examining capability changes driven by intelligent technologies. To our knowledge, this is one of the first studies to conceptualize and operationalize three types of skill adaptation from an individual perspective. Furthermore, our consistent finding across studies that reskilling mitigates deskilling and enables upskilling highlights that adaptation is not a simple substitution-or-enhancement trade-off but an interconnected process of capability reconfiguration.

Second, this study contributes to understanding the GAI productivity paradox by differentiating learning performance and demonstrating how distinct skill adaptation pathways serve as underlying mechanisms explaining the outcomes of GAI dependence. Existing studies present conflicting views on GAI's net impact (e.g., Larson et al., 2024; Lim et al., 2023), partly because performance is often treated monolithically or narrowly defined (Al-Qaysi et al., 2024; Jo, 2023; Zhou & Kim, 2024). This research addresses this gap by distinguishing between routine performance, related to efficiency and quality in standard tasks, and innovative performance, related to creativity and novel strategy generation, and exploring how three types of skill adaptations affect them, offering a new research perspective for the AI-enabled education literature. Furthermore, our qualitative study also provided some potential mechanisms for understanding the impact of students' adaptation on innovative performance, such as freed cognitive resources or broadened knowledge exposure from GAI use, that might mitigate the expected downsides of deskilling for innovation, strongly responding to existing research calls for an in-depth exploration of GAI's impact on students' creativity and critical thinking (Dwivedi et al., 2023; Larson et al., 2024; Skulmowski, 2024). By demonstrating these differential effects, our research provides a mechanism-based explanation for the productivity paradox (Belanche et al., 2024; Flavián et al., 2024): GAI's impact is contingent on the type of skill adaptation fostered and the specific performance dimension examined, enriching the understanding of managing knowledge-focused activities and offering a more insightful perspective than previous studies focusing only on direct usage effects (Abbas et al., 2024; Rehman et al., 2024; Wang & Yin, 2025).

Third, this study contributes to AST in the context of human-GAI interaction by incorporating individual differences, identifying GAI dependence patterns, and investigating how context shapes these patterns. While prior research often overlooks individual differences in GAI use (e.g., Budhathoki et al., 2024; Gao et al., 2024; Jo, 2024) or has failed to provide detailed mechanisms for skill adaptations (Dornelles et al., 2023; Li, 2022; Rinta-Kahila et al., 2023), our research applies and extends AST to the GAI context in several ways (Schmitz et al., 2016). Specifically, we integrate AGT to demonstrate how individual characteristics (i.e., learning goals) act as inputs influencing skill adaptation processes (Nevo et al., 2021), thereby reinforcing AST's principle that agent characteristics shape structuration in human-GAI interaction (Gupta & Bostrom, 2009; Schmitz et al., 2016). Furthermore, we provide concrete ways to analyze the structural consequences of GAI appropriation within an AST framework by conceptualizing and operationalizing deskilling, reskilling, and upskilling (derived from LPT) as distinct outcomes of GAI use. Our finding that reskilling influences deskilling and upskilling supports AST's notion of emergent structures influencing existing ones within the adaptation process (DeSanctis & Poole, 1994; Schmitz et al., 2016). Most importantly, and informed by our qualitative findings, we conceptualize and validate distinct GAI appropriation modes (i.e., SU and AU) using the automation-augmentation framework (Guo et al., 2024; Raisch & Krakowski, 2021), linking AST's appropriation concept directly to the psychological tensions of feeling "empowered vs. replaced" (Flavián et al., 2024). Our experiment then provided causal evidence that task characteristics (analyzability and variety) significantly influence the adoption of these appropriation modes (Crumbly et al., 2025; Daft & Macintosh, 1981; Goodhue & Thompson, 1995). Crucially, these modes led to divergent structuration outcomes via different skill adaptation pathways, thus extending AST by understanding how context shapes appropriation in the GAI context. This perspective goes beyond previous studies that view the process of learning skill changes as technology-driven (Budhathoki et al., 2024; Söllner et al., 2018), emphasizing the importance of human agency in learning from the perspective of a sociotechnological system. In summary, by incorporating both contextual factors (driving appropriation modes) and individual factors (shaping adaptation processes), our study offers a comprehensive AST-based model for understanding adaptation to intelligent technologies like GAI.

8.3. Practical implications

This research carries practical significance in three ways. First, for educators and organizational trainers focused on workforce adaptation, our findings indicate the importance of fostering performance-approach goals among learners and employees. Individuals motivated to demonstrate competence are more likely to engage in beneficial reskilling and upskilling when using GAI, positioning them to feel their capabilities are being augmented rather than substituted by the technology. Recognizing that GAI dependence inherently carries the risk of deskilling, training programs should not only focus on GAI usage but also actively supplement this with foundational skill development and critical thinking exercises to counteract potential skill erosion. Furthermore, the task characteristics framework (Fig. 4) can guide trainers in tailoring usage recommendations, encouraging augmentative use for complex, varied tasks demanding innovation, while managing the risks of substitutive use in simpler, routine contexts.

Second, for practitioners strategizing AI integration, understanding the distinction between substitutive and augmentative use is critical for effective AI deployment. Practitioners should be aware that simply providing access to GAI does not guarantee positive outcomes; the context of use (task type) significantly shapes interaction patterns and subsequent results. Therefore, implementing GAI requires a context-aware strategy. Moreover, policies and work designs should guide employees towards augmentative use where appropriate, and proactively address the potential psychological tension of feeling replaced.

Third, for GAI developers, our findings suggest designing systems that actively facilitate augmentative use. This could involve features that prompt user reflection, encourage iterative interaction, and scaffold complex tasks rather than fully automating them. Designing GAI to genuinely augment users' capabilities, rather than merely substituting their effort, should be a guiding principle. In summary, by understanding the interplay of individual characteristics, task contexts, specific use patterns, and the resulting skill adaptations and psychological experiences (e.g., capabilities being augmented vs. substituted), educators and organizations can develop more effective strategies for AI integration, workforce adaptation, and responsible technology design, thereby maximizing the benefits while minimizing the risks.

8.4. Limitations and future research

Our study is limited in several aspects. First, our empirical investigation was conducted exclusively with university students in China. While this context is highly relevant due to widespread GAI adoption and the presence of diverse tools, findings may not be directly generalizable to other cultural or national contexts. Factors such as differing pedagogical practices, dominant language models, and underlying cultural orientations could influence GAI dependence, skill adaptation processes, and performance outcomes. Therefore, caution is warranted when extrapolating our findings. Future research involving cross-cultural comparative studies will validate and potentially extend our model. In a similar vein, the contextual boundaries of our model could be further explored by examining other crucial task dimensions beyond analyzability and variety, such as task centrality. Second, this study only captures students' perceptions of the impact of GAI tools on their skills changes and learning performance, and does not involve relevant indicators of their actual skills and performance change. Future research could consider using more diverse data to verify the effects of GAI tool use on skill adaptation and learning performance. Third, our research relies on self-reported skill adaptations and use patterns, which, although indicative of underlying cognitive engagement, do not directly measure cognitive adaptation processes. Similarly, consistent with common practice in empirical AST research (Hardin et al., 2014; Liang et al., 2015; Rob et al., 2025; Schmitz et al., 2016; Shao & Li, 2022), our survey captures the stages, not exactly the long-term continuous evolution. Study 1 employed a cross-sectional survey, and Study 3 involved a single-point experimental interaction. While the phrasing of our skill adaptation measures was designed to capture perceived changes over time and our qualitative interviews explored participants' experiences with GAI use evolution, these approaches cannot fully substitute for longitudinal research. Therefore, future research could conduct longitudinal designs to investigate the long-term persistence of skill adaptation.

CRediT authorship contribution statement

Zihan Zeng: Writing – original draft, Validation, Investigation, Formal analysis, Data curation. **Qinwei Li:** Writing – original draft, Validation, Formal analysis. **Bo Yang:** Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Yongqiang Sun:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Study 1

Table A1
Measures.

Constructs	Items	Sources
Deskilling	DS1. Some knowledge can be obtained through GAI tools, so that I no longer need to master this knowledge myself.	Self-developed
	DS2. Some skills can be replaced through GAI tools, so that I no longer need to master these skills myself.	
	DS3. GAI tools make some abilities less important and eliminate the need for me to develop them.	
Upskilling	US1. With GAI tools, I can better understand the relevant knowledge*.	Self-developed
	US2. With GAI tools, I can better master the relevant skills.	
	US3. With GAI tools, I can better develop the relevant abilities.	
Reskilling	RS1. My ability to use GAI tools to obtain relevant knowledge has improved.	Self-developed
	RS2. My ability to use GAI tools to master relevant skills has improved.	
	RS3. My ability to use GAI tools to develop relevant abilities has improved.	
Technology dependence	TD1. GAI tools have become the major tools for my learning.*	Shu et al. (2011)
	TD2. Most of my learning tasks are done with the support of GAI tools.	
	TD3. It would be difficult to imagine my learning without GAI tools.	
	TD4. Most of my learning activities are carried out by GAI tools.	
	TD5. GAI tools have become part of the daily routine in my learning life.	
Performance-avoidance goal	PVG1. My aim is to avoid doing worse than other students.	Elliot and Murayama (2008)
	PVG2. I am striving to avoid performing worse than others.	
	PVG3. My goal is to avoid performing poorly compared to others.	
Performance-approach goal	PPG1. My aim is to perform well relative to other students.	Elliot and Murayama (2008)
	PPG2. I am striving to do well compared to other students.	
	PPG3. My goal is to perform better than the other students.	
Routine performance	RP1. GAI tools help me to accomplish my tasks more quickly.*	Ng et al. (2022)
	RP2. GAI tools improve my learning performance.	
	RP3. GAI tools enhance my academic effectiveness.*	
	RP4. GAI tools help review and eliminate errors in my learning tasks.*	
	RP5. GAI tools help me to realize my future learning target.	
	RP6. GAI tools help me acquire new knowledge.	
Innovative performance	IP1. GAI tools help me to create new ideas for difficult issues.	Janssen (2000)
	IP2. GAI tools help me to search out new learning methods, techniques, or instruments.	
	IP3. GAI tools help me to generate original solutions for problems.	

Table A2
Demographic characteristics (N = 306).

Variables	Levels	Frequency	Percentage (%)
GAI tools	ChatGPT	219	71.6
	ERNIE Bot	87	28.4
Gender	Male	115	37.6
	Female	191	62.4
Age	18 or below	4	1.3
	19–22	256	83.7
	23–29	43	14.1
	30 or above	3	1
Education	Two-year college student	5	1.6
	Undergraduate	280	91.5
	Graduate	21	6.9
Major	Arts and humanities	50	16.3
	Engineering and technologies	133	43.5
	Life science and medicine	36	11.8
	Natural science	17	5.6
	Social science and management	70	22.9

Table A3
Descriptive statistics, reliabilities, and correlations.

	CR	AVE	Mean	SD	DS	IP	PPG	PVG	RP	RS	TD	US
DS	0.930	0.816	3.066	1.510	**0.903**							
IP	0.802	0.574	5.342	0.869	− 0.025	**0.758**						
PPG	0.807	0.584	5.400	0.912	0.060	0.304	**0.764**					
PVG	0.848	0.651	5.009	1.265	0.177	0.016	0.244	**0.807**				
RP	0.767	0.524	5.484	0.773	0.060	0.627	0.422	0.111	**0.724**			
RS	0.780	0.543	5.709	0.724	− 0.087	0.482	0.251	0.066	0.563	**0.737**		
TD	0.840	0.568	4.948	0.964	0.315	0.330	0.267	0.046	0.420	0.299	**0.754**	
US	0.808	0.678	5.698	0.848	− 0.145	0.554	0.308	− 0.052	0.589	0.542	0.326	**0.823**

Note: CR = Composite Reliability, AVE = Average Variance Extracted, SD = Standard Deviation, DS = Deskilling, IP = Innovative Performance, PPG = Performance-approach Goal, PVG = Performance-avoidance Goal, RP = Routine Performance, RS = Reskilling, TD = Technology dependence, US = Upskilling. Diagonal elements are the square roots of AVE.

Table A4
Loadings and cross loadings.

	PVG	PPG	DS	IP	RP	RS	TD	US
PVG1	**0.768**	0.216	0.127	0.025	0.070	0.047	0.041	− 0.020
PVG2	**0.818**	0.205	0.121	0.019	0.123	0.107	0.019	− 0.037
PVG3	**0.833**	0.176	0.175	− 0.001	0.077	0.012	0.051	− 0.064
PPG1	0.119	**0.850**	− 0.009	0.291	0.389	0.234	0.212	0.357
PPG2	0.274	**0.738**	0.141	0.219	0.261	0.143	0.185	0.126
PPG3	0.251	**0.697**	0.066	0.153	0.284	0.175	0.229	0.125
DS1	0.147	0.075	**0.899**	0.025	0.069	− 0.038	0.316	− 0.114
DS2	0.175	0.060	**0.914**	− 0.050	0.062	− 0.118	0.288	− 0.120
DS3	0.157	0.025	**0.897**	− 0.042	0.030	− 0.076	0.244	− 0.164
IP1	0.059	0.213	− 0.026	**0.761**	0.479	0.374	0.293	0.358
IP2	0.013	0.271	− 0.033	**0.713**	0.420	0.341	0.206	0.417
IP3	− 0.027	0.210	− 0.001	**0.797**	0.522	0.381	0.253	0.474
RP2	0.085	0.408	0.036	0.460	**0.743**	0.383	0.376	0.437
RP5	0.047	0.288	0.175	0.435	**0.719**	0.391	0.322	0.382
RP6	0.109	0.223	− 0.075	0.465	**0.709**	0.447	0.217	0.458
RS1	0.026	0.194	− 0.088	0.233	0.294	**0.666**	0.123	0.231
RS2	0.045	0.131	− 0.060	0.385	0.395	**0.735**	0.210	0.434
RS3	0.066	0.230	− 0.056	0.410	0.513	**0.803**	0.290	0.476
TD2	0.059	0.193	0.310	0.300	0.351	0.222	**0.756**	0.236
TD3	− 0.002	0.263	0.243	0.225	0.314	0.163	**0.739**	0.205
TD4	0.053	0.207	0.241	0.275	0.318	0.276	**0.800**	0.259
TD5	0.021	0.150	0.144	0.182	0.281	0.229	**0.719**	0.281
US2	− 0.023	0.245	− 0.106	0.444	0.513	0.421	0.288	**0.822**
US3	− 0.063	0.262	− 0.132	0.468	0.457	0.471	0.249	**0.825**

Note: PVG = Performance-avoidance Goal, PPG = Performance-approach Goal, DS = Deskilling, IP = Innovative Performance, RP = Routine Performance, RS = Reskilling, TD = Technology dependence, US = Upskilling.

Table A5
Common method bias tests (study 1).

Items	Trait factor loading (R1)	$R1^2$	Method factor loading (R2)	$R2^2$
PVG1	0.826	0.682	0.011	0.000
PVG2	0.800	0.640	− 0.019	0.000
PVG3	0.798	0.637	0.008	0.000
PPG1	0.682	0.465	0.166	0.028
PPG2	0.847	0.717	− 0.084	0.007
PPG3	0.811	0.658	− 0.058	0.003
DS1	0.898	0.806	0.045	0.002
DS2	0.906	0.821	− 0.012	0.000
DS3	0.907	0.823	− 0.033	0.001
IP1	0.788	0.621	0.002	0.000
IP2	0.702	0.493	− 0.001	0.000
IP3	0.783	0.613	− 0.001	0.000
RP2	0.763	0.582	0.036	0.001
RP5	0.724	0.524	− 0.021	0.000
RP6	0.683	0.466	− 0.018	0.000
RS1	0.750	0.563	− 0.205	0.042
RS2	0.726	0.527	0.008	0.000
RS3	0.750	0.563	0.181	0.033
TD1	0.596	0.355	− 0.041	0.002
TD2	0.725	0.526	0.070	0.005
TD3	0.756	0.572	− 0.034	0.001
TD4	0.775	0.601	0.024	0.001
TD5	0.691	0.477	− 0.030	0.001
US2	0.823	0.677	0.008	0.000
US3	0.824	0.679	− 0.008	0.000
Average		0.604		0.005

Note: PVG = Performance-avoidance Goal, PPG = Performance-approach Goal, DS = Deskilling, IP = Innovative Performance, RP = Routine Performance, RS = Reskilling, TD = Technology dependence, US = Upskilling.

Appendix B. Study 2

Table B1
Interview questions.

1. What GAI tools do you typically use in your learning?
2. In what types of learning tasks do you choose to use GAI tools?
3. What do you think your learning goals are in the context of your studies?
4. How dependent do you feel on GAI tools during your learning process? Do you feel that your reliance on this technology has increased?
5. Has the use of GAI tools affected your learning methods, abilities, or skills? What impacts have you observed?
6. Do you believe that using GAI tools has improved your learning performance? How has it affected you?
7. In the process of using GAI tools, do you feel that some of the learning skills, methods, or abilities you previously had have been enhanced? What impact has this change had on your learning performance?
8. During the use of GAI tools, did you proactively learn how to make the tools perform better for your learning tasks and integrate these newly acquired skills into your learning activities? What impact has this change had on your learning performance?
9. Do you feel that some of your skills can be replaced by GAI tools, making it unnecessary for you to master these skills, methods, or abilities yourself? What impact has this change had on your learning performance?
10. What role do you think GAI tools play in your learning process?
11. Do you think the way you use GAI tools differs across different learning tasks?
12. Do you think the performance of GAI tools influences how you use them and your learning outcomes?
13. What overall impact do you think GAI tools have had on your learning? What are the positive and negative impacts? Are you concerned that GAI tools might lead to a decline in some of your personal abilities?
14. Is there anything else you would like to add regarding this interview?

Table B2
Demographic profile of interview respondents.

(Resp #)	Gender	Age	GAI Tools Usage Exp. (yrs)	GAI Tools Usage Frequency	Most Used GAI Tools	Second Most Used Generative AI tools	Grade Level	Major
R1	Female	21	1–2	Every day	ChatGPT	Newbing	Undergraduate year 4	Social science and management
R2	Female	22	0.5–1	Once a week	ChatGPT	Doubao	Undergraduate year 4	Social science and management
R3	Female	24	1–2	Once a week	ChatGPT	Doubao	PhD student	Engineering and technologies

(continued on next page)

Table B2 (*continued*)

(Resp #)	Gender	Age	GAI Tools Usage Exp. (yrs)	GAI Tools Usage Frequency	Most Used GAI Tools	Second Most Used Generative AI tools	Grade Level	Major
R4	Female	21	1–2	Once every two weeks	ChatGPT	ERNIE Bot	Undergraduate year 4	Engineering and technologies
R5	Male	26	1–2	Once a week	Perplexity	ChatGPT	PhD student	Life science and medicine
R6	Female	21	1–2	Every day	ChatGPT	Kimi	Undergraduate year 4	Social science and management
R7	Female	20	1–2	Once a week	Kimi	ERNIE Bot	Undergraduate year 3	Social science and management
R8	Female	22	1–2	Once a week	ChatGPT	ERNIE Bot	Undergraduate year 4	Social science and management
R9	Male	26	1–2	Every day	ChatGPT	ChatGPT	PhD student	Engineering and technologies
R10	Male	20	1–2	Every day	ChatGPT	ERNIE Bot	Undergraduate year 4	Social science and management
R11	Female	22	0.5–1	Once a month	ERNIE Bot	ChatGLM	Undergraduate year 4	Engineering and technologies
R12	Male	26	1–2	Every day	ChatGPT	Doubao	PhD student	Social science and management
R13	Female	20	1–2	Once a week	ChatGPT	ERNIE Bot	Undergraduate year 3	Arts and humanities
R14	Female	21	1–2	Once a week	ERNIE Bot	ChatGPT	Undergraduate year 3	Arts and humanities
R15	Male	19	1–2	Every day	ChatGPT	Kimi	Undergraduate year 3	Social science and management
R16	Female	21	1–2	Once every two weeks	ChatGPT	Kimi	Undergraduate year 3	Arts and humanities

Table B3
Illustrative examples of coding for qualitative analysis.

Responses	Initial Coding		Consensus Coding
	Coder 1	Coder 2	
"When I receive a writing task, the **first thing that comes to mind** is to use ChatGPT to generate text, then I modify the text generated by ChatGPT, instead of writing it myself." (R3, female, PhD student, engineering and technologies).	TD(+)	TD(+)	TD(+)
"Sometimes, I try feeding the entire English article to GAI tools to have it generate a Chinese summary directly, which makes reading literature **highly efficient**. I don't need to read the article in detail myself to grasp the information, and the summary generated by GAI tools is much **better than my own reading**." (R16, female, undergraduate, arts and humanities).	RP(+)	RP(+)	RP(+)
"The content generated by GAI tools sometimes gives me **inspiration** and adds **new perspectives** to my original ideas." (R8, female, undergraduate, social science and management).	IP(+)	IP(+)	IP(+)
"GAI tools cannot provide me with good ideas. I believe good ideas still come from reading literature and communicating with peers. Although the specific steps to implement these ideas can be handled by GAI tools to **improve task efficiency**, I don't believe they offer significant help in terms of **innovation**." (R12, male, PhD student, social science and management).	RP(+)	RP(+)	RP(+)
"During the process of using GAI tools to assist me with academic writing tasks in English, I feel that my writing ability has **indeed improved**, but I also feel that my writing ability **seems to have declined** (if there were no GAI tools). Although it has increased my **writing efficiency,** I am unsure about the extent to which my effort and ability influence the writing results generated by GAI." (R9, male, PhD student, engineering and technologies).	DS(+) RP(+) US(+)	TD(+) DS(+) RP(+) US(+)	DS(+) RP(+) US(+)
"GAI tools are very convenient and **have replaced much of my work**. For example, when writing papers in the past, I would search and integrate information myself, but now I directly ask GAI tools to provide me with the results. This method has somewhat reduced my need to think about the problem and **has replaced my search capabilities**." (R13, female, undergraduate, arts and humanities).	TD(+)→DS(+)	DS(+)	TD(+)→DS(+)
"During the use of GAI tools, in order to achieve the desired results, I **ask it from different angles**, **learn prompt writing**, and **consciously improve my prompt skills.**" (R2, female, undergraduate, social science and management).	TD(+)→RS(+)	RS(+)	TD(+)→RS(+)
"I feel that my ability in programming tasks **has improved**. I used to only program in Python, but now with the help of GAI tools, **I am able to write more difficult Java code**. Moreover, after reviewing the code generated by GAI tools, I have **deepened my understanding** and mastery of programming logic." (R10, male, undergraduate, social science and management).	TD(+)→US(+)	TD(+)→US(+)	TD(+)→US(+)
"For **those less important writing tasks**, I rely on GAI tools to help me complete them. I only need to evaluate the generated output, **without having to write it myself**." (R11, female, undergraduate, engineering and technologies).	PVG(+)→DS(+)	PVG(+)→DS(+)	PVG(+)→DS(+)
"Once GAI tools help me complete the task, **the course requirements are met,** and I **no longer go deeper** to explore or improve." (R13, female, undergraduate, arts and humanities).	PVG(+)→US(-)	PVG(+)→US(-)	PVG(+)→US(-)
"With **the goal of performing better**, I **repeatedly adjust my instructions** to make the content generated by GAI tools continuously improve in order to achieve my desired outcome." (R6, female, undergraduate, social science and management).	PPG(+)→RS(+)	PPG(+)→RS(+)	PPG(+)→RS(+)
"**I asked GAI tools to propose a research idea**, and through this process, I **gained a deeper understanding** of research paradigms suitable for similar research contexts. In other words, with the help of GAI tools, I am becoming more familiar with this research field." (R7, female, undergraduate, social science and management).	PPG(+)→US(+)	PPG(+)→US(+)	PPG(+)→US(+)
"In order to make better use of GAI tools, I work hard to **develop my ability to write prompts**. With clear prompt instructions, the output from GAI tools improves, so I use it more effectively to **enhance my literature reading skills.**" (R1, female, undergraduate, social science and management).	RS(+)→US(+)	RS(+)→US(+)	RS(+)→US(+)
"**My proactive optimization of how I use GAI tools** means I need to have a more comprehensive understanding of my learning tasks. At the same time, improving the effectiveness of GAI tools requires me to **clarify my task ideas**, so I believe enhancing my skills in using GAI tools can **prevent a decline in my own abilities.**" (R7, female, undergraduate, social science and management).	RS(+)→DS(-)	RS(+)→DS(-) RS(+)→US(+)	RS(+)→DS(-)

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Table B3 (*continued*)

Responses	Initial Coding		Consensus Coding
	Coder 1	Coder 2	
"**With the help of GAI tools**, my paper writing **speed has increased**, and it has **saved my energy and attention**." (R9, male, PhD student, engineering and technologies).	DS(+)→RP (+)	DS(+)→RP (+)	DS(+)→RP(+)
"I have **now learned to open a new chat window for each question** to avoid being influenced by the historical chat records. Additionally, I **provide some background data** so GAI tools can better learn my task context. This approach improves the quality of its responses, increasing both accuracy and relevance, helping me **complete tasks effectively**." (R14, female, undergraduate, arts and humanities).	RS(+)→RP (+)	RS(+)→RP (+)	RS(+)→RP(+)
"**Exploratory use** of GAI tools can help me **generate some ideas** I hadn't thought of before while writing experimental reflections." (R11, female, undergraduate, engineering and technologies).	RS(+)→IP (+)	RS(+)→IP (+)	RS(+)→IP(+)
"Using GAI tools to complete my programming learning tasks **improved my coding skills**, significantly **enhanced the efficiency** of completing assignments, and ultimately led to excellent grades on my submissions." (R8, female, undergraduate, social science and management).	US(+)→IP (+)	US(+)→IP (+) RS(+)→IP (+)	US(+)→IP(+)
"If I have an innovative idea, GAI tools will help me **better organize my thoughts**, making the idea **more refined**." (R1, female, undergraduate, social science and management).	US(+)→IP (+)	US(+)→IP (+)	US(+)→IP(+)
"GAI tools help me save a considerable amount of time and effort on routine tasks, and the time and energy saved can be used for innovative tasks. Additionally, GAI tools allow me to **gain and enhance knowledge in various fields**, and based on this knowledge and skill enhancement, I will have **more innovative inspiration**." (R5, male, PhD student, life science and medicine).	US(+)→IP (+) RCR(+)	US(+)→IP (+)	US(+)→IP(+)
"When completing creative tasks, I need GAI tools to replace me in handling some routine tasks, such as organizing language, to **release my cognitive load**. I don't need to spend so much energy on those routine tasks, thus freeing up more energy and focus for **exploring more creative tasks**." (R1, female, undergraduate, social science and management).	RCR(+)→IP (+)	RCR(+)→IP (+)	RCR(+)→IP(+)
"GAI tools are trained on extensive corpora, so the texts generated by GAI tools contain professional vocabulary or fields that I have not encountered before. They can cross professional barriers and **teach me content from other fields** relevant to my tasks, giving me a **broader perspective and more inspiration**. Compared to my own writing, the content generated by GAI tools may be more creative." (R6, female, undergraduate, social science and management).	EKD(+)→IP (+)	EKD(+)→IP (+)	EKD(+)→IP(+)

Note: PVG = Performance-avoidance Goal, PPG = Performance-approach Goal, DS = Deskilling, RS = Reskilling, US = Upskilling, TD = Technology dependence, IP = Innovative Performance, RP = Routine Performance, RCR = Release Cognitive Resources, EKD = Expand Knowledge Domains; → implies cause-effect relationship.

Qualitative Analysis Procedure

To ensure the methodological rigor of our qualitative study, we followed the established guidelines for sequential mixed-methods research (e.g., Lu et al., 2024; Srivastava & Chandra, 2018). The qualitative phase was designed with the dual purpose of corroboration/confirmation and complementarity. To achieve this, we ensured both design validity and analytical validity.

Design Validity, which concerns the credibility of the study design and execution, was ensured through the rigorous selection of 16 participants who were experienced GAI users from diverse academic backgrounds (see Table B2), providing a rich and relevant sample. Furthermore, our semi-structured interview protocol (see Table B1) was designed to be open-ended, allowing participants the freedom to communicate their thoughts and experiences comprehensively.

Analytical Validity, which pertains to the rigor of data analysis and interpretation, was maintained through a systematic coding process. We employed a multiple classification scheme (Srivastava & Chandra, 2018), initially informed by the core constructs of our quantitative research model. The analysis proceeded in the following steps:

1. Independent coding: Two authors independently coded the full set of interview transcripts using NVivo 12. Each response segment could be assigned to one or more categories based on the coding scheme.
2. Consensus and refinement: The coders then met regularly to compare their codes, discuss interpretations, and resolve any discrepancies through discussion until a consensus was reached. The valence (positive or negative) of statements was noted, along with any divergent responses that did not fit the initial framework.
3. Thematic aggregation: This process involved structuring the data by moving from specific, participant-centric concepts (first-order codes) to more abstract conceptual categories (second-order themes), and finally to the overarching theoretical dimensions (aggregate dimensions).

Table B4 presents our final data structure.

Table B4
Data structure and coding scheme for qualitative analysis.

Aggregate Dimensions	Second-Order Themes	First-Order Codes (Illustrative Examples from Data)
Technology Dependence	Pervasive Reliance on GAI	"When faced with a task, my first instinct is to try asking ChatGPT" (R2). "I've replaced much of my work with GAI tools" (R15). "I dump any task onto GAI tools to handle" (R12).
Skill Adaptation	Deskilling (Skill Substitution) Reskilling (Skill Expansion) Upskilling (Skill Complementarity)	"Replaced my coding capabilities" (R4). "Ability for writing has decreased" (R3). "Consciously improve my prompt skills" (R10). "Repeatedly adjust my instructions" (R3). "Strengthened my programming thinking" (R10). "My writing and translation skills have been significantly improved" (R2). "Enhance my coding skills" (R1).
Learning Goal	Performance-Avoidance Goal	"For those less important writing tasks, my goal is just to meet a basic standard" (R10). "Course requirements are met, and I submit it directly" (R1).
	Performance-Approach Goal	"With the goal of achieving better grades" (R2). "Striving to perform excellently" (R14).

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Table B4 (*continued*)

Aggregate Dimensions	Second-Order Themes	First-Order Codes (Illustrative Examples from Data)
Learning Performance	Routine Performance	"Improved my writing productivity and accuracy" (R3). "Enhanced my learning efficiency" (R15).
	Innovative Performance	"Brought me many new perspectives and new ways to solve problems" (R2). "Help me generate some new ideas" (R4).

Appendix C. Study 3 – Online experiment details

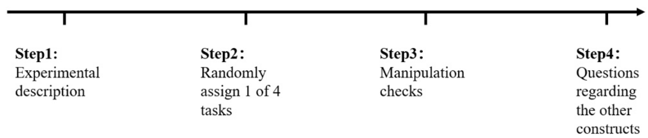

Fig. C1. Scenario experiment procedure.

Table C1
Scenario experiment materials.

Manipulation Description	Manipulation Utilized
Unanalyzable and low-variety tasks (N = 99)	**[Scenario Introduction]** Below is a description of a category of learning tasks you might encounter when using GAI tools. **[Manipulation of Task Characteristics]** This category of learning tasks is characterized by the following features: (1) The core value of the task lies in originality and subjective judgment, and there are no pre-defined answers. The process for completing these tasks is difficult to break down, and it relies on intuition, experience, and creative thinking. (2) The task content is relatively fixed, with little variation. Although the task requires creativity, the scope of the problem is well-defined, and there are fewer unexpected situations. **[Task Examples for Participants]** Next, we will provide you with three examples of such tasks (in your actual studies, tasks may include but are not limited to these examples) to help you understand their characteristics and recall similar situations you have actually experienced. Please read the following three scenario descriptions carefully for at least 30 s until you understand the characteristics of this type of task. 1. This semester, in a general education course on innovative thinking, your instructor asks you to conceive a core creative idea for a compelling slogan for a brand-new, conceptual eco-friendly product. There is no standard answer for this task; the key is whether your idea is appealing and aligns with the product's philosophy. The assessment is primarily based on the uniqueness of your creative concept. Although the specific product might differ each time, this type of creative ideation task will occur regularly this semester. 2. This semester, in an ethics or public policy discussion class you are taking, the instructor regularly assigns tasks that require you to express your viewpoint on social topics. While these topics (e.g., AI ethics) are repeatedly covered in class, your task is to propose your own unique, in-depth arguments, rather than searching for standard answers. The evaluation focuses mainly on the persuasiveness of your argumentation or the originality of your thinking. This type of argument construction task will appear frequently this semester. 3. This semester, as a member of a research group focused on green development, you are periodically required to brainstorm novel preliminary solutions for specific environmental issues (e.g., energy conservation on campus, reducing plastic use). There is no single correct solution for these tasks; the emphasis is on the innovativeness of the ideas and their alignment with sustainability concepts. The evaluation is primarily based on the quality of your creative ideas. Although the specific problem context may vary slightly, this kind of innovative ideation within a familiar domain will be a recurring activity.
Unanalyzable and high-variety tasks (N = 99)	**[Scenario Introduction]** Below is a description of a category of learning tasks you might encounter when using GAI tools. **[Manipulation of Task Characteristics]** This category of learning tasks is characterized by the following features: (1) The tasks are open-ended and unpredictable, requiring flexible adaptation. These tasks often involve many exceptions or novel situations that cannot be predicted in advance. (2) The core value of the task lies in originality and subjective judgment, and there are no pre-defined answers. The process for completing these tasks is difficult to break down, and it relies on intuition, experience, and creative thinking. **[Task Examples for Participants]** Next, we will provide you with three examples of such tasks (in your actual studies, tasks may include but are not limited to these examples) to help you understand their characteristics and recall similar situations you have actually experienced. Please read the following three scenario descriptions carefully for at least 30 s until you understand the characteristics of this type of task. 1. This semester, for a major course's final project (or your thesis), you need to write a detailed research proposal. This requires you to identify a valuable research question on your own, support your arguments by extensively reviewing diverse and even conflicting literature, and design a feasible research methodology that you must explore independently. Throughout this process, you often cannot foresee the specific challenges you will face, which may require you to adjust your research direction, integrate varied information from multiple sources, and make many critical decisions at any time. The assessment of this task is not based on a standard answer but focuses on the overall quality, innovativeness, and persuasiveness of your research proposal. 2. This semester, you are participating in a cutting-edge, cross-disciplinary engineering design project or scientific research topic (e.g., developing a novel biosensor or designing a smart city solution). The project's goal is ambitious and may be vaguely defined initially, requiring you to integrate knowledge from different disciplines, attempt to use new technologies that may not be fully mature, and solve technical and integration challenges that continuously emerge and are difficult to predict. The project's path is not linear; it requires you and your team to constantly explore, trial, and iterate, making critical technical and design decisions. The evaluation will be based on the innovation level of the final outcome, its functionality, and its effectiveness in solving the problem, rather than on following a set procedure. 3. This semester, you need to independently complete a small-scale social survey report. You are required to choose your own topic, focusing on an

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Table C1 (*continued*)

Manipulation Description	Manipulation Utilized
	issue that is underexplored or controversial. You must broadly collect diverse, and even conflicting, literature or data, construct your own analytical framework, and ultimately form a persuasive and unique academic argument. The entire process is exploratory. You may face unforeseen challenges such as difficulties in data acquisition, obstacles in theoretical application, or the need for major revisions to your arguments, requiring constant judgment and integration. The final evaluation is primarily based on the research's innovativeness, depth of argumentation, and academic rigor, rather than adherence to a fixed model.
Analyzable and low-variety tasks (N = 100)	**[Scenario Introduction]** Below is a description of a category of learning tasks you might encounter when using GAI tools. **[Manipulation of Task Characteristics]** This category of learning tasks is characterized by the following features: (1) The task content is relatively routine and highly repetitive. When performing such tasks, the number and frequency of abnormal, unexpected, or novel events are comparatively low, meaning the process and potential issues demonstrate high predictability with lower uncertainty. (2) There are clear execution steps or objective criteria. These tasks can largely be broken down into several well-defined components, with distinct, established rules and procedures available for guidance throughout the process. **[Task Examples for Participants]** Next, we will provide you with three examples of such tasks (in your actual studies, tasks may include but are not limited to these examples) to help you understand their characteristics and recall similar situations you have actually experienced. Please read the following three scenario descriptions carefully for at least 30 s until you understand the characteristics of this type of task. 1. This semester, one of your courses requires you to read key paragraphs from an important English academic article and translate them accurately and fluently into Chinese. This task has very clear evaluation criteria, focusing on translation accuracy (fidelity to the original text) and fluency (adherence to Chinese expression conventions). You need to ensure precise terminology and correct grammar. Typically, the requirements and steps for this type of translation task are very similar, and you will need to complete it multiple times this semester. 2. This semester, several of your courses require that submitted reports or papers must follow a specific, very strict academic formatting guide (e.g., APA, MLA, or a specific journal format). The guide has clear rules for every detail. You need to ensure your document fully complies with these formatting standards. This formatting and layout work, with its clear and consistent rules, is a task you frequently need to handle. 3. This semester, in one of your lab courses, you are required to write lab reports multiple times based on standard experiments you have completed. These types of reports usually have very clear structural requirements and formatting specifications. You need to follow these established rules to objectively and accurately document your experimental process and conduct basic analysis. The evaluation is primarily based on the completeness and clarity of your report, as well as your strict adherence to the formatting requirements. This standardized report writing is a recurring, routine requirement for this type of course.
Analyzable and high-variety tasks (N = 99)	**[Scenario Introduction]** Below is a description of a category of learning tasks you might encounter when using GAI tools. **[Manipulation of Task Characteristics]** This category of learning tasks is characterized by the following features: (1) The task content is varied, and the problems and situations encountered are different. The execution of the task involves a high frequency of unexpected events and novel situations. (2) The tasks have clear rules but involve subjective interpretation or unexpected complexities. Although there are established procedures or standards, their application requires significant experience and judgment to handle various ambiguous or exceptional cases. **[Task Examples for Participants]** Next, we will provide you with three examples of such tasks (in your actual studies, tasks may include but are not limited to these examples) to help you understand their characteristics and recall similar situations you have actually experienced. Please read the following three scenario descriptions carefully for at least 30 s until you understand the characteristics of this type of task. 1. This semester, a major course project requires you to conduct a systematic literature review, and you must strictly adhere to an extremely detailed set of operational procedures provided by your advisor. These procedures specify which databases you must use, what precise screening criteria to apply, and which standardized tables to use for information extraction. Although the rules are very clear, you find that challenges are endless in practice: when screening articles, you always encounter cases where the criteria are vaguely defined; when extracting information from papers with varying formats, it is often difficult to find all the necessary and precise content. 2. This semester, you need to use a very detailed evaluation guide with clear judgment criteria to grade a large batch of course assignments submitted by your classmates (e.g., course essays, project reports). This guide provides clear proficiency-level descriptions and scoring standards for each evaluation dimension (such as clarity of argument, evidentiary support, and logical structure). However, the specific writing style, strengths, and weaknesses of each assignment vary greatly, requiring you to carefully and consistently apply the clear scoring criteria to every assignment of different style and quality. 3. This semester, in one of your courses, you learned how to apply a highly structured, step-by-step content analysis method (e.g., frame analysis for news reports or thematic coding for interview transcripts). The instructor provides a detailed codebook or analysis guide with clear and specific rules. Your task is to apply this method to analyze a large volume of real-world news reports or interview transcripts with diverse content. In practice, you find that although the coding rules are explicit, the specific phrasing, implicit meanings, and ambiguous areas in each text are endless, requiring you to repeatedly and carefully apply the rules to each unique and specific text segment.

Table C2
Measures.

Constructs	Items	Sources
Manipulation checks: task analyzability	TA1. This task in my learning is guided by clear standards, rules, and procedures. TA2. In completing this learning task, there is an understandable sequence of steps that can be followed. TA3. In this type of learning task, students actually rely on established methods and procedures. TA4. Existing materials (e.g., standards, textbooks, tutorials) are available to guide students through the task.	(Song & Sun, 2018)
Manipulation checks: task variety	TV1. There is variety in the problems that arise during this task. TV2. The challenges encountered in this learning task are dissimilar from one instance to another. TV3. When a problem arises, it takes a lot of experience and training to know what to do. TV4. This learning task can be described as non-routine.	(Song & Sun, 2018)
Scenario realism	How realistic do you think this type of learning task is in actual learning? (1 = absolutely not realistic, 7 = very realistic)	(Lowry et al., 2017)

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Table C2 (*continued*)

Constructs	Items	Sources
Technology dependence: Substitutive use	SU1. I hand over this task to GAI with little or no further involvement. SU2. GAI can autonomously complete this task without an option for me to intervene. SU3. I have difficulties learning from the task execution process of GAI. SU4. GAI entails replacing my role with advanced AI technologies. SU5. I have a limited understanding of GAI's rationale and thinking.* SU6. I have limited perceptions of accountability for the GAI's outputs.*	(Guo et al., 2024)
Technology dependence: Augmentative use	AU1. I remain active during the task completion process. AU2. I collaborate closely with GAI to accomplish the learning task. AU3. The use of the GAI complements my capabilities (e.g., knowledge absorption, critical thinking) rather than lessens or restricts them. AU4. I play an important role in compensating for GAI errors.* AU5. GAI tool provides decision support for me in completing the task.*	(Guo et al., 2024)

Note: The measures of Deskilling, Innovative Performance, Performance-approach Goal, Performance-avoidance Goal, Routine Performance, Reskilling, and Upskilling in Study 3 were consistent with those used in Study 1.

Table C3
Demographic characteristics (N = 397).

Variables	Levels	Frequency	Percentage (%)
GAI tools	DeepSeek	181	45.6
	Doubao	102	25.7
	ChatGPT	47	11.8
	ERNIE Bot	31	7.8
	Kimi	27	6.8
	Others	9	2.3
Age	Under 18	1	0.3
	18–20	116	29.2
	21–23	204	51.4
	24–26	68	17.1
	27 and older	8	2.1
Education	Associate college	27	6.8
	Undergraduate	272	68.5
	Master	87	21.9
	PhD student	11	2.8
Major	Arts and humanities	55	13.9
	Engineering and technologies	110	27.7
	Life science and medicine	53	13.4
	Natural science	18	4.5
	Social science and management	161	40.6
Experience	Under 3 months	19	4.8
	3–6 months	56	14.1
	6–12 months	103	25.9
	1–2 years	184	46.3
	2 years and above	35	8.8
Frequency	At least once a day	2	0.5
	At least once every three days	12	3.0
	At least once a week	16	4.0
	At least once every two weeks	86	21.7
	At least once a month	166	41.8
	Less than once a month	115	29.0

Appendix D. Study 3 – Results

Table D1
Descriptive statistics, reliabilities, and correlations.

	CR	AVE	Mean	SD	AU	DS	IP	PPG	PVG	RP	RS	SU	US
AU	0.810	0.587	5.302	0.923	**0.766**								
DS	0.927	0.809	2.859	1.374	− 0.248	**0.899**							
IP	0.873	0.695	5.481	0.876	0.423	− 0.185	**0.834**						
PPG	0.941	0.842	5.082	1.215	0.247	− 0.060	0.209	**0.918**					
PVG	0.908	0.766	5.181	1.245	0.062	0.020	0.051	0.293	**0.875**				
RP	0.777	0.537	5.438	0.754	0.516	− 0.175	0.505	0.284	0.113	**0.733**			
RS	0.866	0.683	5.416	0.838	0.511	− 0.217	0.468	0.176	0.102	0.529	**0.826**		
SU	0.869	0.624	3.068	1.164	− 0.492	0.289	− 0.258	− 0.068	0.059	− 0.287	− 0.324	**0.790**	
US	0.887	0.723	5.433	0.943	0.626	− 0.319	0.526	0.232	0.060	0.600	0.606	− 0.437	**0.850**

Note: CR = Composite Reliability, AVE = Average Variance Extracted, SD = Standard Deviation, AU = Augmentative Use, DS = Deskilling, IP = Innovative Performance, PPG = Performance-approach Goal, PVG = Performance-avoidance Goal, RP = Routine Performance, RS = Reskilling, SU = Substitutive Use, US = Upskilling. Diagonal elements are the square roots of AVE.

Table D2
Loadings and cross loadings.

	AU	DS	IP	PPG	PVG	RP	RS	SU	US
AU1	**0.786**	− 0.234	0.322	0.226	− 0.016	0.405	0.344	− 0.371	0.487
AU2	**0.766**	− 0.156	0.299	0.184	0.131	0.428	0.424	− 0.331	0.463
AU3	**0.746**	− 0.180	0.349	0.158	0.027	0.353	0.404	− 0.428	0.488
DS1	− 0.241	**0.898**	− 0.161	− 0.047	0.010	− 0.165	− 0.202	0.265	− 0.290
DS2	− 0.199	**0.922**	− 0.168	− 0.018	0.023	− 0.142	− 0.193	0.252	− 0.264
DS3	− 0.227	**0.877**	− 0.170	− 0.094	0.022	− 0.164	− 0.190	0.261	− 0.304
IP1	0.341	− 0.127	**0.840**	0.199	0.052	0.401	0.372	− 0.206	0.436
IP2	0.246	− 0.110	**0.827**	0.119	0.055	0.394	0.334	− 0.170	0.393
IP3	0.448	− 0.213	**0.835**	0.196	0.024	0.461	0.451	− 0.259	0.477
PPG1	0.206	− 0.054	0.169	**0.905**	0.286	0.263	0.154	− 0.057	0.196
PPG2	0.253	− 0.049	0.217	**0.921**	0.253	0.239	0.180	− 0.066	0.230
PPG3	0.219	− 0.061	0.186	**0.926**	0.271	0.283	0.149	− 0.063	0.209
PVG1	0.011	0.038	0.016	0.250	**0.847**	0.049	0.065	0.055	− 0.017
PVG2	0.115	0.040	0.055	0.228	**0.869**	0.079	0.080	0.058	0.059
PVG3	0.024	− 0.008	0.049	0.287	**0.909**	0.137	0.109	0.046	0.077
RP2	0.352	− 0.077	0.324	0.220	0.159	**0.693**	0.367	− 0.142	0.364
RP5	0.340	− 0.054	0.300	0.219	0.094	**0.723**	0.351	− 0.157	0.383
RP6	0.432	− 0.223	0.463	0.195	0.020	**0.781**	0.435	− 0.302	0.544
RS1	0.401	− 0.124	0.404	0.201	0.080	0.472	**0.816**	− 0.280	0.487
RS2	0.420	− 0.188	0.354	0.128	0.097	0.353	**0.844**	− 0.258	0.474
RS3	0.442	− 0.224	0.398	0.108	0.078	0.475	**0.818**	− 0.264	0.535
SU1	− 0.409	0.214	− 0.154	− 0.089	− 0.009	− 0.184	− 0.286	**0.828**	− 0.348
SU2	− 0.316	0.206	− 0.126	− 0.004	0.016	− 0.137	− 0.166	**0.764**	− 0.241
SU3	− 0.407	0.246	− 0.317	− 0.060	0.065	− 0.325	− 0.339	**0.768**	− 0.429
SU4	− 0.400	0.239	− 0.169	− 0.044	0.111	− 0.212	− 0.181	**0.800**	− 0.310
US1	0.541	− 0.200	0.461	0.217	0.045	0.482	0.488	− 0.356	**0.806**
US2	0.522	− 0.297	0.419	0.173	0.031	0.488	0.498	− 0.37	**0.867**
US3	0.534	− 0.313	0.461	0.200	0.075	0.556	0.557	− 0.386	**0.876**

Note: AU = Augmentative Use, DS = Deskilling, IP = Innovative Performance, PPG = Performance-approach Goal, PVG = Performance-avoidance Goal, RP = Routine Performance, RS = Reskilling, SU = Substitutive Use, US = Upskilling.

Table D3
The effects of control variables (study 3).

Testing Study 1 Hypotheses in Study 3	Path coefficients (t-value)
H1a: PVG→DS	$0.028^{n.s.}$ (t = 0.476)
H1b: PPG→DS (-)	$-0.013^{n.s.}$ (t = 0.231)
H2a: PVG→RS (-)	$0.070^{n.s.}$ (t = 1.216)
H2b: PPG→RS	$0.039^{n.s.}$ (t = 0.747)
H3a: PVG→US (-)	$-0.013^{n.s.}$ (t = 0.248)
H3b: PPG→US	0.074* (t = 1.760)

Note: DS = Deskilling, RS = Reskilling, US = Upskilling, PVG = Performance-avoidance Goal, PPG = Performance-approach Goal, *$p < 0.1$, n.s. = not significant.

Table D4
Common method bias tests (study 3).

Items	Trait factor loading (R1)	$R1^2$	Method factor loading (R2)	$R2^2$
AU1	0.797	0.635	− 0.046	0.002
AU2	0.768	0.590	− 0.025	0.001
AU3	0.732	0.536	0.074	0.005
DS1	0.897	0.805	− 0.009	0.000
DS2	0.928	0.861	0.034	0.001
DS3	0.872	0.760	− 0.026	0.001
IP1	0.848	0.719	− 0.025	0.001
IP2	0.850	0.723	− 0.137	0.019
IP3	0.806	0.650	0.166	0.028
PPG1	0.911	0.830	− 0.014	0.000
PPG2	0.913	0.834	0.019	0.000
PPG3	0.930	0.865	− 0.004	0.000
PVG1	0.903	0.815	− 0.043	0.002
PVG2	0.869	0.755	0.012	0.000
PVG3	0.871	0.759	0.033	0.001
RP2	0.729	0.531	− 0.095	0.009
RP5	0.762	0.581	− 0.154	0.024
RP6	0.717	0.514	0.248	0.062
RS1	0.817	0.667	0.023	0.001
RS2	0.856	0.733	− 0.101	0.010

(*continued on next page*)

Table D4 (*continued*)

Items	Trait factor loading (R1)	$R1^2$	Method factor loading (R2)	$R2^2$
RS3	0.806	0.650	0.078	0.006
SU1	0.843	0.711	0.010	0.000
SU2	0.813	0.661	0.145	0.021
SU3	0.694	0.482	− 0.224	0.050
SU4	0.822	0.676	0.048	0.002
US1	0.800	0.640	0.051	0.003
US2	0.874	0.764	− 0.105	0.011
US3	0.875	0.766	0.055	0.003
Average		0.697		0.009

Note: AU = Augmentative Use, DS = Deskilling, IP = Innovative Performance, PPG = Performance-approach Goal, PVG = Performance-avoidance Goal, RP = Routine Performance, RS = Reskilling, SU = Substitutive Use, US = Upskilling.

Data availability

Data will be made available on request.

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